

PROJECT REPORT – 5

1. Project title and summary

SRM Institute of Science and Technology
College of Engineering and Technology
Department of ECE

AY 2018-2019
15EC496L -Major Project Details

Sl No	Register No	Students Name(s)	Project Supervisor	Project Title
1	RA1511004010107	Pooja Anand	Dr. P. Eswaran	Low Cost Digitalization (Industry 4.0) Solution for Siemens Sinumerik CNC System to Increase the Transparency and Utilization of the Machine.
	RA1511004010059	Vinitha Lea Philip		
2	RA1511004010553	kedar prasad karpe	DR. R .Kumar	Scalable Cooperative Transport Strategy Using A Group Of Simple Robots
	RA1511004010511	Dhruv pant		
	RA1511004010712	Nimish pastaria		
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SRM Institute of Science & Technology
College of Engineering and Technology
Department of Electronics and Communication Engineering
Project Summary - 2018-2019

Sl No	Students Name	Project Guide	Project Title	Objective of the Project	Realistic constraints imposed	Standards to be referred/followed	Multidisciplinary tasks involved	Outcome
1	VINITHA LEA PHILIP [Reg No:RA1511004010059] POOJA ANAND [Reg No:RA1511004010107]	Dr. P. Eswaran	Low Cost Digitalization (Industry 4.0) Solution for Siemens Sinumerik CNC System to Increase the Transparency and Utilization of the Machine	To build a solution that is economical, can be adopted by small and medium enterprises so that they get a taste of how IoT can be adopted by monitoring some of the critical machine parameters thereby trying to reduce or prevent breakdowns of machines and associated productivity.	Safe place has to be found in each machine to place the KIT & Adaptors, so that no damage happens to the kit or connecting cables, by machine operator or any other activity on the machine. Power fluctuations or failures may affect the normal working of the kit.	Open user interface design based on WinCC or Run MyHMI Up to 10 machining channels per NCU The digital twin – end-to-end development and new business models	1) Electrical and Electronics Engineering for utilizing Raspberry pi 2) Computational and IT field for programming the raspberry pi using Python 3) Desktop publication for report.	Journal Publication SNIP:0.354

2	KEDAR PRASAD KARPE [Reg No: RA1511004010553] DHRUV PANT [Reg No: RA1511004010511] JAYATI SINGH [Reg No: RA1511004010654] NIMISH PASTARIA [Reg No: RA1511004010712]	Dr R. Kumar	SCALABLE COOPERATIVE TRANSPORT STRATEGY USING A GROUP OF SIMPLE ROBOTS	The develop a solution to the cooperative transport problem is based on an articulated drive model where the group of robots has leader and multiple follower robots.	Calibration Sensors: Registration Modeling NIST will produce robot models, datasets, software tools, and calibration artifacts that can lead to easily calibrated or even self-calibrating sensors and robots.	ISO 10218-1:2011	Microcontroller architecture for handling communication and motor control SPRINTER robot which represents all the separate components of the mechanical design	IEEE Conference (Malaysia) IEEE Conference (Brasil)
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PROJECT REPORT – 5

2. Project report

**COST EFFECTIVE DIGITALIZATION SOLUTION
FOR SIEMENS SINUMERIK CNC SYSTEM TO
INCREASE THE TRANSPARENCY AND
UTILIZATION OF THE MACHINE**

A PROJECT REPORT

Submitted by

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Under the guidance of

Dr. P.ESWARAN, Ph.D

(Associate Professor, Department of Electronics and Communication & Engineering)

in partial fulfillment for the award of the degree

of

BACHELOR OF TECHNOLOGY

in

**ELECTRONICS AND COMMUNICATION
ENGINEERING**

of

FACULTY OF ENGINEERING AND TECHNOLOGY



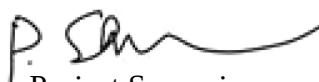
S.R.M. Nagar, Kattankulathur, Kancheepuram District

MAY 2019

BONAFIDE CERTIFICATE

Certified that this project report titled **“COST EFFECTIVE DIGITALIZATION SOLUTION FOR SIEMENS SINUMERIK CNC SYSTEM TO INCREASE THE TRANSPARENCY AND UTILIZATION OF THE MACHINE”** is the bonafide work of **“VINITHA LEA PHILIP [RA1511004010059], POOJA ANAND [RA1511004010107]”** who carried out the project work under my supervision as a batch. Certified further, that to the best of my knowledge the work reported herein does not form any other project report on the basis of which a degree or award was conferred on an earlier occasion for this or any other candidate.

Date :


Project Supervisor


Head of the Department

Submitted for University Examination held on _____ in _____ the
Department of Electronics and Communication Engineering, SRM Institute of Science
and Technology, Kattankulathur.

Date:

Internal Examiner

External Examiner

DECLARATION

We hereby declare that the Major Project entitled “**COST EFFECTIVE DIGITALIZATION SOLUTION FOR SIEMENS SINUMERIK CNC SYSTEM TO INCREASE THE TRANSPARENCY AND UTILIZATION OF THE MACHINE**” to be submitted for the Degree of Bachelor of Technology is our original work as a team and the dissertation has not formed the basis of any degree, diploma, associateship or fellowship of similar other titles. It has not been submitted to any other University or institution for the award of any degree or diploma.

Place: Chennai

Date:

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ABSTRACT

In this work, we are bringing the current trends in the field of data exchange in manufacturing to the Computer Numerical Control (CNC) machine in a cost-effective manner. To achieve this, we program in Python language in a modern board (raspberry pi). The parameters to be monitored are preselected. Using this as a guide the program is written in Python language. Display screens have been designed using Graphical user interface (GUI) which helps the user to analyze the machine in real time. The objective of the project is to reduce downtime, thereby increasing efficiency and in turn profitability.

We are aiming at building a solution that is economical, can be adopted by small and medium enterprises so that they get a taste of how IoT can be adopted to do a successful predictive maintenance by monitoring some of the critical machine parameters thereby trying to reduce or prevent breakdowns of machines and associated productivity. It will be a simple solution that will not require any special knowledge or skill in using it so it can be adopted by a larger section of the industry.

ACKNOWLEDGEMENTS

We would like to express our deepest gratitude to our Founder Chancellor **Dr. T. R. Paarivendhar**, Chairman **Mr. Ravi Pachamoothoo**, President **Dr. P. Sathyanarayanan** for providing us the necessary facilities for the successful completion of our course.

We also would like to acknowledge our Vice Chancellor **Dr. Sandeep Sancheti**, ProVice Chancellor **Dr. T. P. Ganesan** and Registrar **Dr. N. Sethuramn** for their constant support and endorsement through invaluable administration. In the same breath, we would also like to mention our sincere gratitude to the Director **Dr. C. Muthamizhchelvan** for his constant support and encouragement.

We would like to express our deepest gratitude to **Dr. T. Rama Rao**, HOD and **Dr.K.VJAYAN**, Project Coordinator for giving us an opportunity to take up this project. We also would like to thank our guide **Dr. P. ESWARAN**, Associate Professor, Department of Electronics and Communication Engineering, SRMIST, Kattankulathur for his valuable guidance, consistent encouragement, personal caring, timely help and providing us with an excellent atmosphere for doing research. All through the work, in spite of his busy schedule, he has extended cheerful and cordial support to us for completing this research work. We would like to express our gratitude towards ELECTRONICS AND COMMUNICATION ENGINEERING DEPARTMENT for giving us an opportunity and encouragement which helped us in completion of this project.

We would like to express our gratitude towards Mr.K.K.Vivek (Service Operations Team Leader (CHN & CBE)) Siemens Ltd, Mr.Joseph S (Vertical (Indirect) Sales Professional (CHN)) Siemens Ltd for giving us an opportunity to work at Siemens Ltd and for their kind co-operation and encouragement which helped us in completion of this project. Our thanks and appreciations to all who have willingly helped us out with their abilities during this period. We would like to extend our sincere thanks to all of them.

Vinitha Lea Philip

Pooja Anand

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ABBREVIATIONS

CNC	Computer Nmerical Control
GPIO	General Purpose Input/Output
IoT	Internet of Things
I/O	Input/Output

CHAPTER 1

INTRODUCTION

1.1 Industry 4.0

. Industry 4.0 is the fourth industrial revolution. It focuses on cyber physical systems, the Internet of Things, cloud computing and cognitive computing. This brings fourth smart factories to the industrial world. Industry 4.0 has four design principles which include: Interconnection, Information transparency, Technical assistance and Decentralized decisions. In our project we will be focusing mainly on the information transparency. Transparency is one of the key aspects as it allows the operators to take well informed decisions based on the data provided. Thus aiding functionality and helping the operators identify the key areas for improvement and thereby increase the utilization. Large and successful companies will easily implement cloud based solution. Small (proprietor) type companies, cannot afford such high initial cost. So we have decided to create a affordable solution for such companies using hardwiring.

1.2 Brief Description of Project

With the IoT gaining importance, this has become one of the most important use cases for the Industry 4.0. The IoT has made information easily available and accessible. For many small business owners, the adoption of IoT may seem like a daunting challenge. In reality, there are many ways small businesses can take advantage of IoT right now. The solutions that are available today are not really economically viable for small and medium scale enterprises. These systems require very expensive and time-consuming machine integrations, with software that is difficult to use. This prevents them from adopting these methods because their return on investment takes a longer time. Our mission is to help manufacturers increase production efficiency (availability, performance, and quality) by simplifying machine monitoring.

We need to make small and medium enterprises to embrace IoT in a big way. These enterprises look for the following: cost should be reasonable, the technology should be

easy to use without any specialist knowledge or having to hire someone with special skills, it should be easily available, and the results should be accurate and must help them save or recover money faster. Our project is aimed to develop a solution that will help small and medium businesses achieve the above objectives.

1.3 Literature Survey

1.3.1 An Industry 4.0-enabled Low Cost Predictive Maintenance

Approach for SMEs

Sezer et al. (2018) outlines the base concepts, materials and methods used to develop an Industry 4.0 architecture focused on predictive maintenance, while relying on low-cost principles to be affordable by Small Manufacturing Enterprises. In this paper, they have developed a low-cost, easy-to-develop system architecture that measures the temperature and vibration variables of a machining process in a Haas CNC turning centre, while storing such data in the cloud.

1.3.2 Real-Time Manufacturing Machine and System Performance

Monitoring Using Internet of Things

Saez et al. (2018) uses a real-time hybrid simulation of manufacturing at a machine and system level. Data from both the virtual and real environments are merged to assess performance. Deviations from expected values represent an error that can trigger a warning signal to production, maintenance, and/or manufacturing personnel at the plant regarding health and productivity of plant operations.

1.3.3 Development of a Cloud-Computing-based Equipment

Monitoring System for Machine Tool Industry

Hung et al. (2012) presents the design of a cloud computing-based equipment monitoring system, called CCEMS, for the CNC machine tool industry. The Graphical User Interface (GUI) plays an important role in the CCEMS. It allows users to interact with the system for controlling and operating equipment. It monitors the performance and

statuses, detecting and diagnosing equipment faults, conjecturing production quality and precision of equipment.

1.3.4 Investigated Information Data of CNC Machine Tool for Established Productivity of Industry 4.0

Chang and Wu (2016) discusses on how the controller tuning operation can change the information data of a CNC machine tool. In this way established productivity of industry 4.0 is investigated. A cloud network is provided to give connectivity to the responses of tuning operation. This helps in share the big data, to support decision making, and to adjust operations in real time. Thus it helps in checking the CNC machine tool for smart productivity based on its tuning operations.

1.3.5 Smart Factories in Industry 4.0: A Review of the Concept and of Energy Management Approached in Production Based on the Internet of Things Paradigm

Shrouf et al. (2014) gives a complete understanding of the interaction between smart factories and customers of Industry 4.0. It provides information about behaviour of both the customers and the products and the characteristics of smart factories. It deals with an approach for improving IoT based energy management in smart factories. It focuses on energy consumption and efficiency along with production management.

CHAPTER 2

METHODOLOGY

2.1 Hardware Design

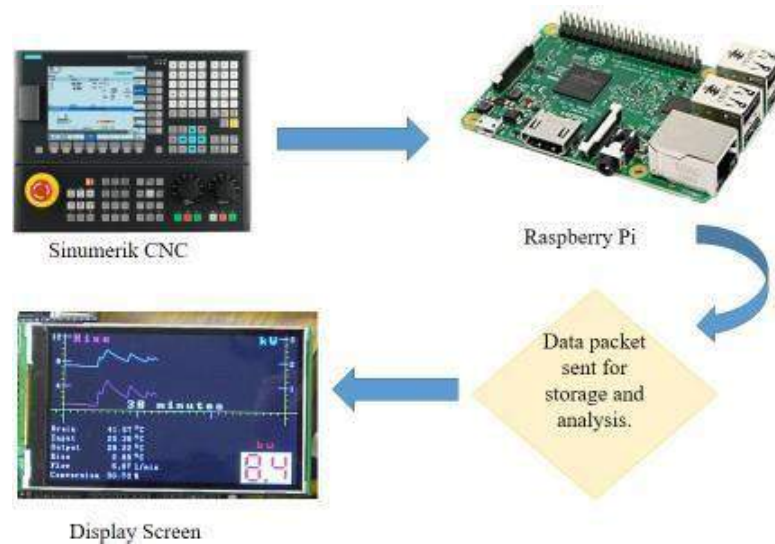


Figure 2.1: System Flow

In our project, we will be monitoring the Sinumerik CNC with the help of a raspberry pi and thereby showing the machines utilization patterns. The process of monitoring is first started by selecting the list of parameters to be monitored. Once the parameters are selected, we program the raspberry pi. The raspberry pi is programmed with the help of the programming language python with respect to the requirements. Then the raspberry pi is hardwired to the Sinumerik CNC in order to collect the required data over a specified time period. After the data is collected by the raspberry pi we collate and display the data. The data is viewed on display screens in the form of graphical representations. Thus this method helps in providing better transparency and offers a platform for development for the small industries in this field.

2.1.1 Raspberry Pi 3

The entire project centres around the Raspberry pi. We will be monitoring the Sinumerik CNC with the help of a raspberry pi. The version we are using is Raspberry pi 3 Model B. It is economical and has built-in wireless connectivity. It is a mini computer that runs on Linux platform and provides us with GPIO (General Purpose Input/Output) pins. It has many ports which can be used with ease. We can easily connect and control electronic components for useful computing. We are using Python language to code the raspberry pi.

2.1.2 Sinumerik CNC

A CNC (Computer Numerical Control) is a device used for material removal to get desired parts/components. The Sinumerik CNC 828D is basically the NC Kernel with a built-in PLC in the front which is connected to an I/O card. In manual control, the operators have to physically prompt the required commands of tools via buttons, leavers and wheels. All these limitations are overcome with the help of the CNC. On activating the CNC, the program starts executing and the desired cuts are performed by the corresponding tools which carry out the tasks like a robot. The part program outlines the placement of the tool in the CNC. This can be used to control many complex machinery including mills, lathes and grinders.

2.1.3 I/O Module

The Sinumerik CNC 828D is basically the NC Kernel with a built-in PLC in the front which is connected to an I/O card. The various parameters that have to be monitored are taken as output from the CNC and given as input to the pi via the I/O card. The I/O module is used to connect digital and analog inputs/outputs. The SINUMERIK I/O Module is PP 72/48D 2/2A PN. It has 72 digital inputs and 48 digital outputs. The digital output is connected to a relay board.

2.1.4 Relay

In our project we need to connect the pi to a module with a higher voltage. Relays are used to avoid the risk of the raspberry pi burning out. The raspberry pi can only handle up to 5V and the GPIOs can tolerate only 3.3V without relays. Relays have two main contacts NO and NC.

Normally Open Contact (NO) - It closes the loop when the relay is switched on and breaks the loop when relay is switched off.

Normally Closed Contact (NC) - It opens the loop when the relay is switched on and is hence known as the break contact. It disconnects the circuit when the relay is inactive.

2.2 Software Design

We use the softwares python 3, PySimpleGUI and tkinter for programming the raspberry pi to create screens for GUI and real time dashboard. We have created a realtime dashboard to provide the live status of the production status. The real time screen has a history button which when selected displays the past data of the machine parameters.

2.2.1 Python 3

Python is designed in such a way that it is highly readable. Python is processed by the interpreter at runtime. The program does not need to be compiled before executing it.

2.2.2 PySimpleGUI

PySimpleGUI helps in solving the GUI challenges by providing a super-simple, easy to understand interface to GUIs that can be easily customized. The PySimpleGUI is being used in our project to make the user interface screens.

2.2.3 Tkinter

Tkinter is a toolkit for Python's GUI package. It is an object-oriented layer. It helps in designing GUI application like widgets for creating user interface screens.

2.3 Parameters

We have decided to monitor the following parameters:

2.3.1 Machine Operating Mode: Auto/Manual

The CNC machine acts like a standard machine in manual mode. When the machine is in manual mode the operator can push buttons, turn wheels, and turn switches on or off. In Auto mode, we execute our program. It allows us to see the commands executed as they happen.

2.3.2 Part Program Running: Yes/No

The set of instruction by which we can produce a part is known as part program and we can check the CNC program.

2.3.3 Cycle Time

The time taken to finish a production run by the amount of fine work pieces produced. Small size businesses benefit most from reductions in Setup time while large size businesses benefit most from reductions in Cycle time.

2.3.4 Part Count

The number of parts that have been produced. It is monitored only when it runs in auto mode.

2.3.5 Feedrate Override

The feed-rate override commonly ranges from 0 to 200 percent. It is a multiposition switch. It enables the setup person to slow (or stop) cutting motions on one end of the spectrum and double the programmed feed rate on the other.

2.3.6 Spindle Running Time

The spindle running hours is defined as the percentage of available time that a machines spindle is on. The related custom macro program is executed whenever the spindle is turned on(commanded by M03 or M04).

2.3.7 Breakdown Hours

The breakdown hour is the amount of time when a system is unavailable or of time that a system fails to perform its primary functions. A breakdown can occur when the equipment stops its functioning due to power loss.

2.3.8 Machine Running Hours

The machine running hours is the working of a machine for an hour. This is used as a basis for cost finding and for determining operating effectiveness.

2.3.9 Machine Ready Time

Time taken after the part is produced till the next part is loaded onto the machine. It tells the worker that the machine is available to start the operation. Depends on the skill of the operator.

2.3.10 Machine Utilization Hours

It is the amount of time the machine is used successfully. Machine utilization compares the run time to the amount of time taken to setup the machine.

CHAPTER 3

IMPLEMENTATION

3.1 Experimental Setup

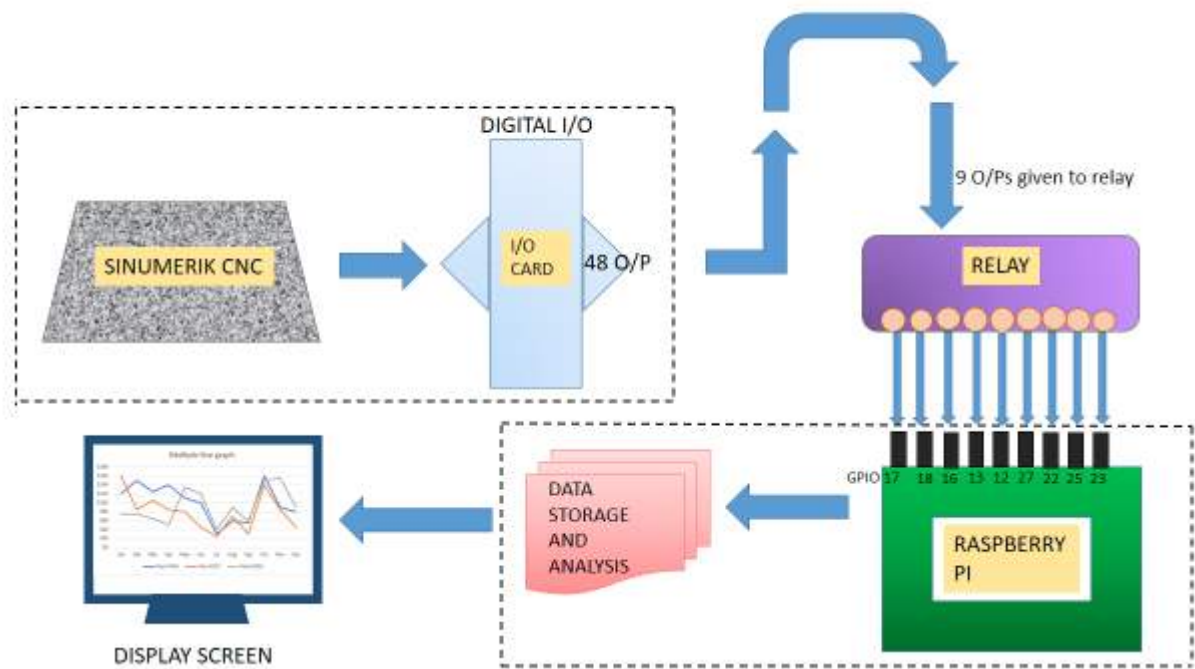


Figure 3.1: Design of Cost Effective Module

The Sinumerik CNC 828D is basically a monitor in the front which is connected to an I/O card. The I/O module is used to connect digital and analog inputs/outputs. The SINUMERIK I/O Module is PP 72/48D 2/2A PN. It has 72 digital inputs and 48 digital outputs. The digital output is connected to a relay board. Relays are used to avoid the risk of the raspberry pi burning out. The raspberry pi can only handle up to 5V and the GPIOs can tolerate only 3.3V without relays. The various parameters that have to be monitored are taken as output from the CNC and given as input to the pi via the I/O card. The python program in the pi will run the proper algorithm to collect and store the data. The data will then be analyzed and displayed graphically for the user to interpret the results easily.

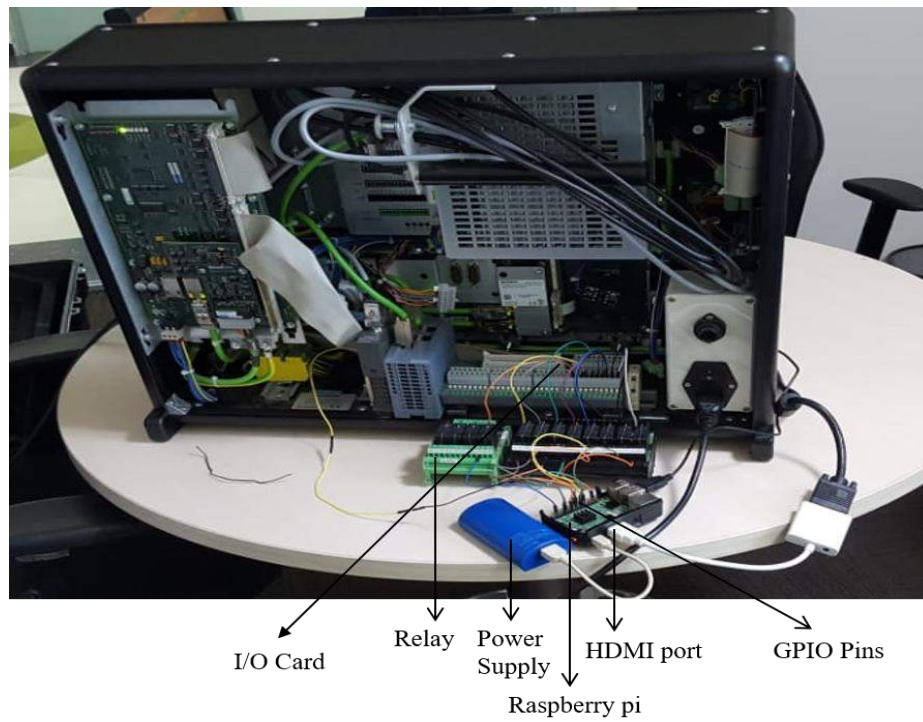


Figure 3.2: Experimental Setup 1

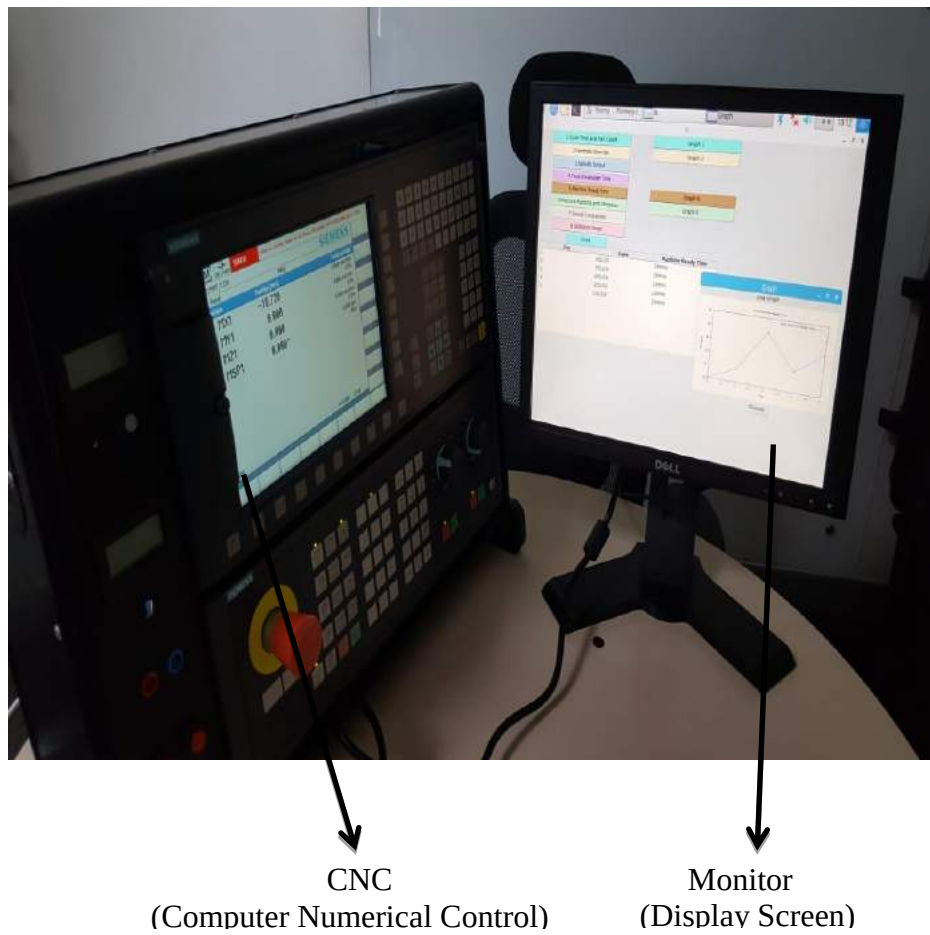


Figure 3.3: Experimental Setup 2

3.2 Program logic for parameters

We are monitoring ten parameters of the CNC, machine operating mode: auto/manual, part program running: yes/no, cycle time, part count, feedrate override, spindle running hours, breakdown hours, machine running hours, ready for operation, machine utilization hours. The flow diagram for these parameters is given below.

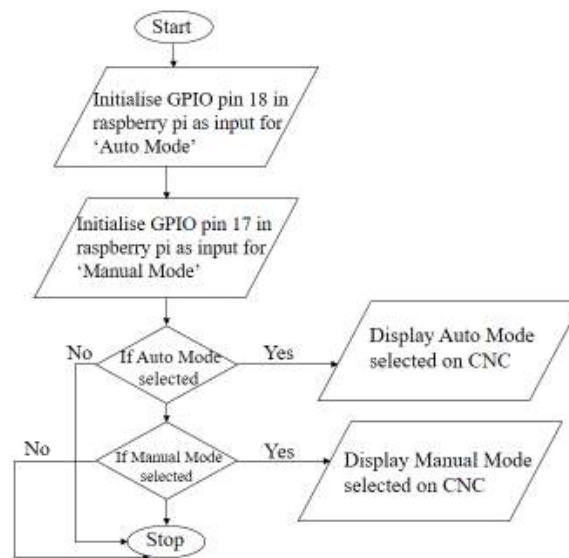


Figure 3.4: Machine Operating Mode: Auto/Manual

Figure 3.4 describes the process of identifying whether the CNC is in auto mode or manual mode

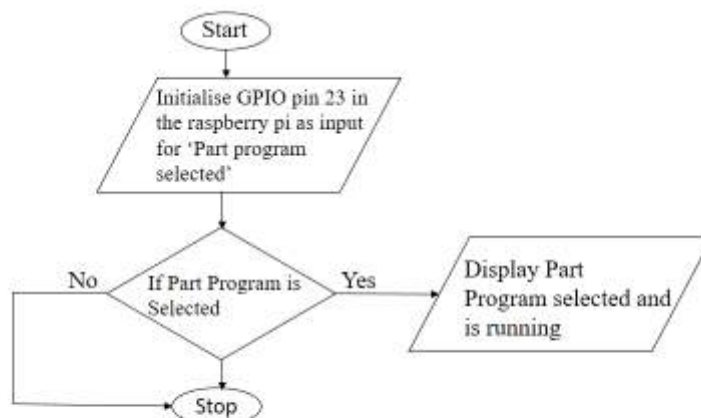


Figure 3.5: Part Program Running: Yes/No

Figure 3.5 depicts how the system is able to identify that the part program is running.

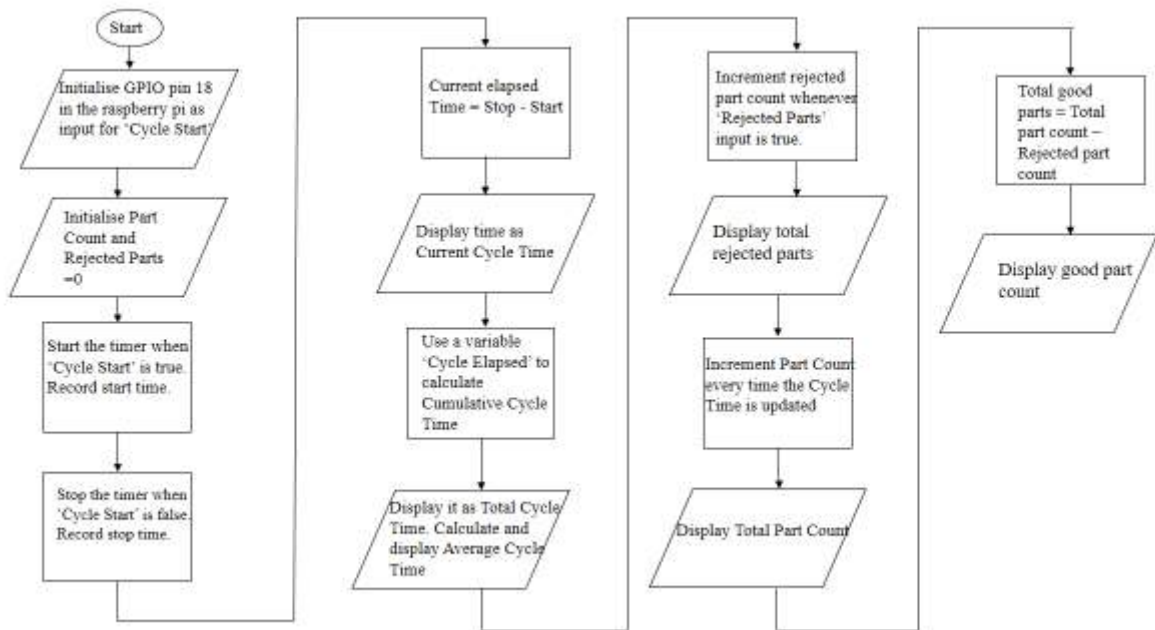


Figure 3.6: Cycle Time

The calculation of current cycle time for every part manufactured the number of good parts and rejected parts, the average and total cycle time is outlined in Figure 3.6.

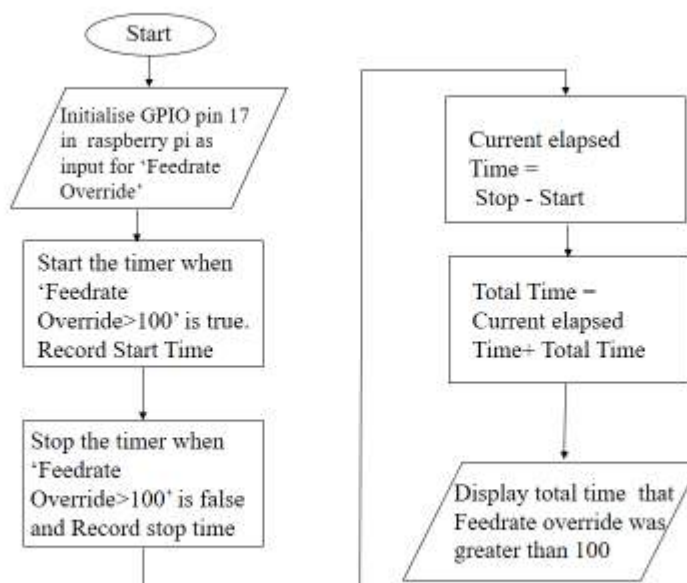


Figure 3.7: Feedrate Override

The time during which federate override are greater than 100 is recorded and displayed as shown in Figure 3.7.

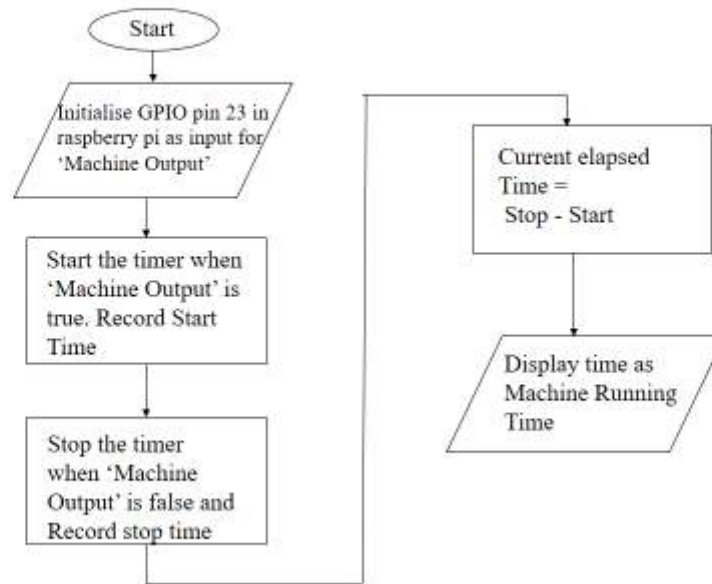


Figure 3.8: Machine Running Hours

Figure 3.8 demonstrates how the total machine running time is calculated

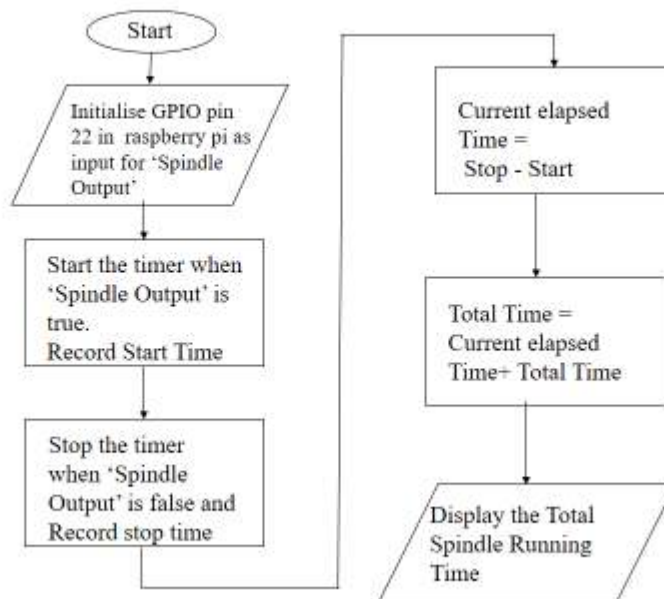


Figure 3.9: Spindle Running Time

Figure 3.9 depicts how the total spindle running time is recorded and displayed.

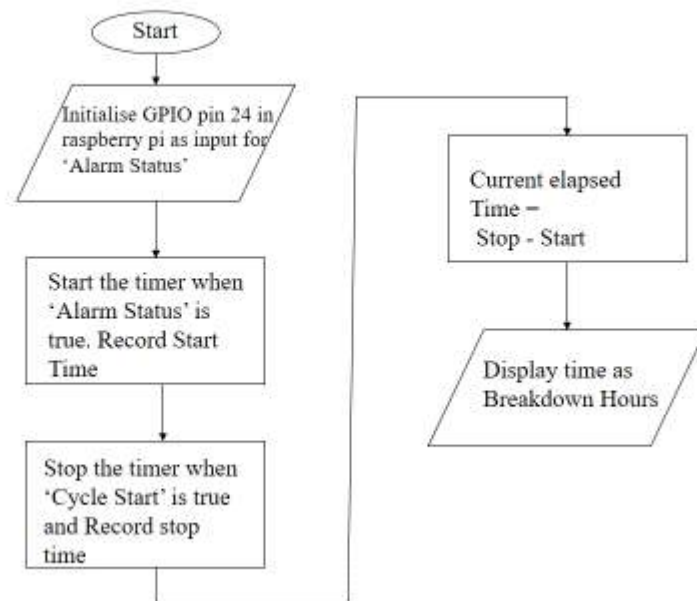


Figure 3.10: Breakdown Hours

The total breakdown time is calculated as shown in Figure 3.10.

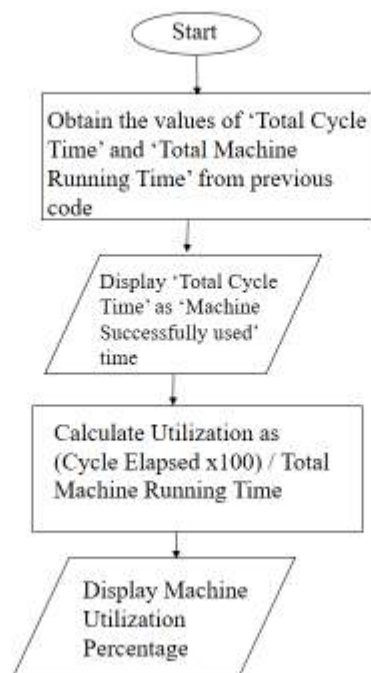


Figure 3.11: Machine Utilization Hours

The machine utilization percentage is calculated as shown in Figure 3.11.

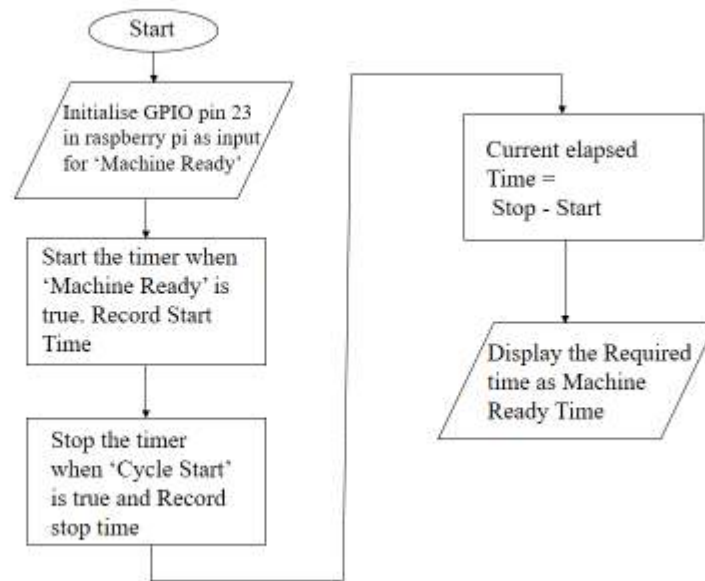


Figure 3.12: Machine Ready

The total time that the CNC was in ready state is calculated as depicted by Figure 3.12.

CHAPTER 4

RESULTS AND DISCUSSION

4.1 Machine Utilization Dashboard

This is a system to aid the information flow regarding the status of the production system. These processes can send a signal or warning if something is wrong. This information is then shown through lights, numbers, and graphics to alert others about the problems. It makes the production status of the machine at current time easily viewable and clear to everyone. It also helps the operator to analyze regularly and ensure production.

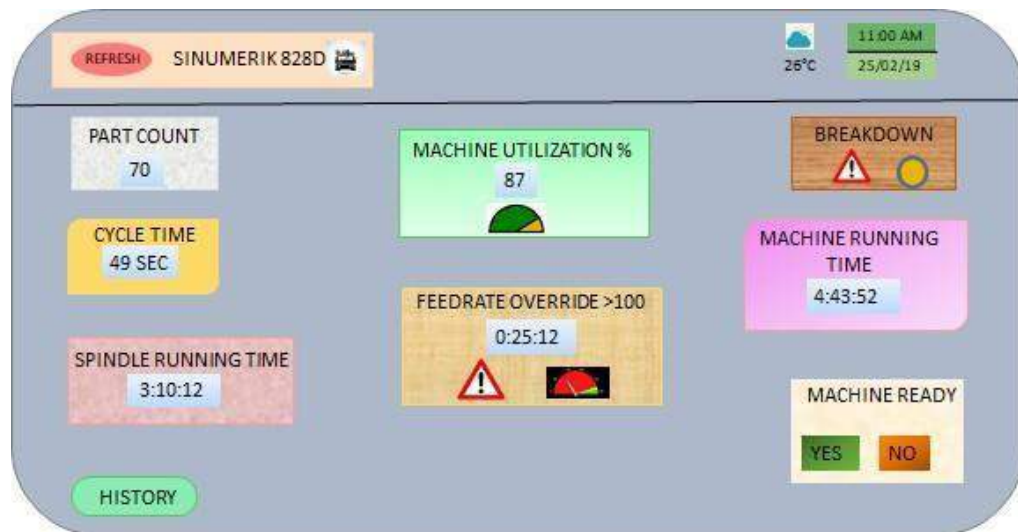


Figure 4.1: Machine Utilization Dashboard

4.2 GUI Display Screens

When the history button in the dashboard is clicked, it shows screens that display the past data as required. The first screen displays the months. On selecting January, February and March, the respective weeks are displayed. The list of parameters will be displayed on selecting the weeks. On selecting each parameter, the corresponding table will be displayed. The display screens are shown below.

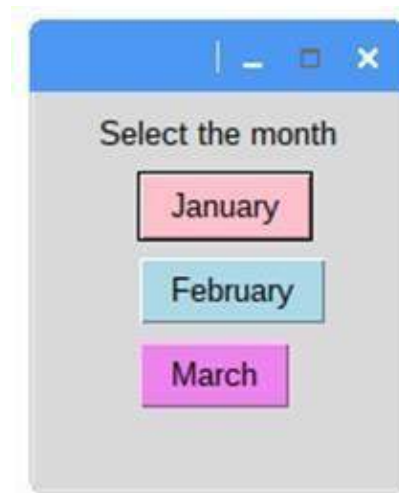


Figure 4.2: Months

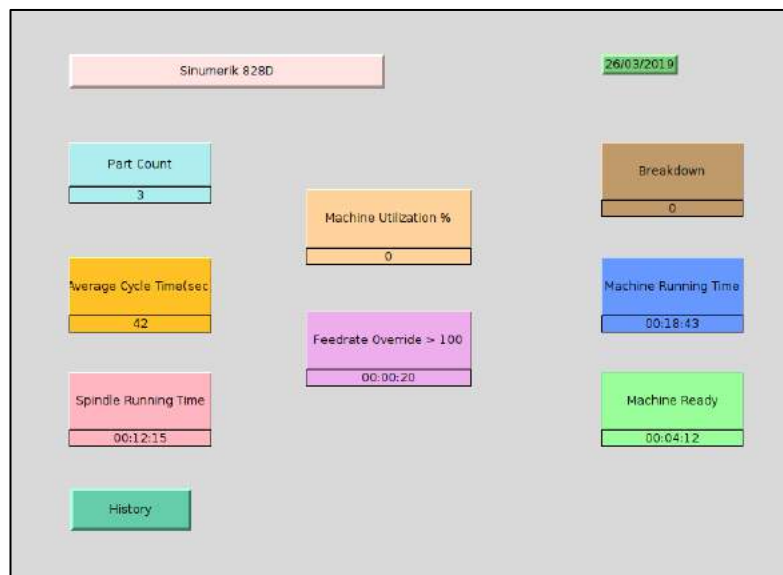


Figure 4.3: Machine Utilization Dashboard



Figure 4.4: Weeks in January



Figure 4.5: Weeks in February



Figure 4.6: Weeks in March

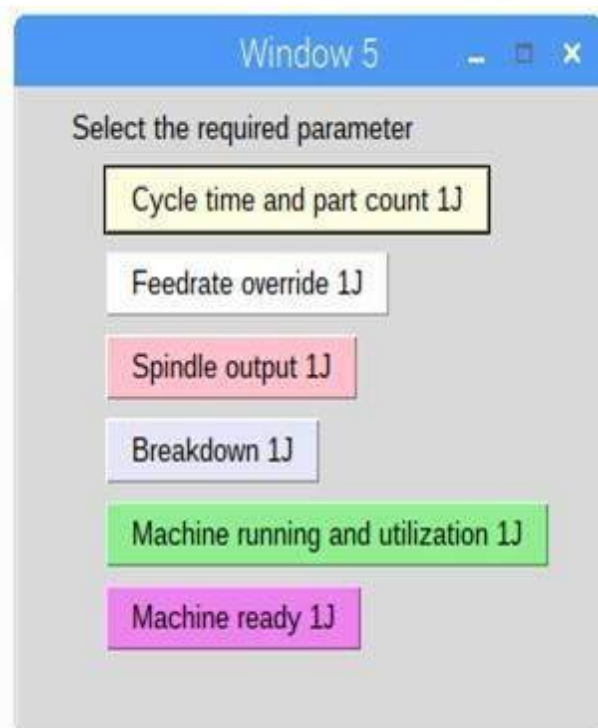


Figure 4.7: Parameters

tk #2				
Day	Date	Average Cycle Time	Total Cycle Time	Part Count
1	1/01/19	48sec	5hr 40min 0sec	425
2	2/01/19	50sec	5hr 45min 0sec	414
3	3/01/19	48sec	5hr 40min 48sec	426
4	4/01/19	51sec	5hr 48min 30sec	410
5	7/01/19	49sec	5hr 43min 49sec	421

Figure 4.8: Cycle time and Part Count

It is inferred from Figure 3.20 that the average time for 1 part to be produced in a day is 48 sec. The total number of parts produced in a day is 425 parts and the total time for 425 parts to be produced is 5hrs 40mins 0sec.

tk #2		
Day	Date	Feedrate Override >100
1	1/01/19	25min
2	2/01/19	55min
3	3/01/19	50min
4	4/01/19	45min
5	7/01/19	20min

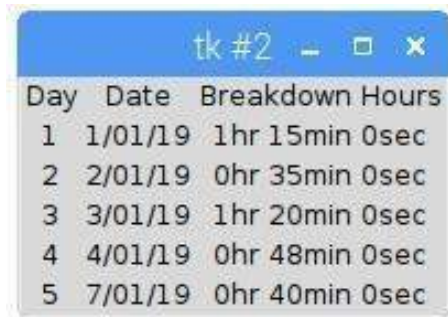
Figure 4.9: Feedrate Override

Figure 3.21 depicts that the feed rate override was greater than 100 for 25mins on 1st January 19. This shows that the operator has increased the programmed feed rate over 100 for 25mins.

tk		
Day	Date	Total Spindle Running Hours
1	1/01/19	6hr 20min 0sec
2	2/01/19	6hr 25min 0sec
3	3/01/19	6hr 20min 48sec
4	4/01/19	6hr 28min 30sec
5	7/01/19	6hr 23min 49sec

Figure 4.10: Spindle Running Time

Figure 3.22 shows the amount of time the spindle has functioned for each day in a week. The total spindle running hours for 1 day is 5hrs 80mins 0sec.



Day	Date	Breakdown Hours
1	1/01/19	1hr 15min 0sec
2	2/01/19	0hr 35min 0sec
3	3/01/19	1hr 20min 0sec
4	4/01/19	0hr 48min 0sec
5	7/01/19	0hr 40min 0sec

Figure 4.11: Breakdown Time

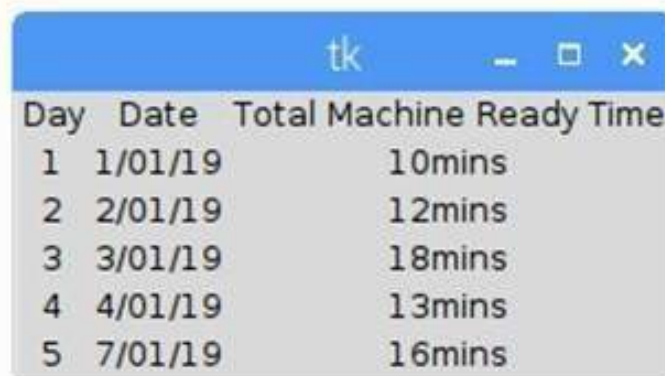
It can be inferred from Figure 3.23 the amount of time in a day when the machine was unavailable or failed to perform its functions. The total breakdown time in a day is 1hr 15min 0sec.



Day	Date	Total Machine Running Time	Machine Utilization Percentage
1	1/01/19	6hr 45min 10sec	86.3 %
2	2/01/19	6hr 46min 18sec	85.4 %
3	3/01/19	6hr 51min 23sec	87.3 %
4	4/01/19	6hr 47min 28sec	82.6 %
5	7/01/19	6hr 43min 16sec	85.1 %

Figure 4.12: Machine Running and Utilization Percentage

The total machine running time in a day was found to be 6hr 45min 10sec from Figure 3.24. The total machine utilization percentage for 1 day is 86.3%.



Day	Date	Total Machine Ready Time
1	1/01/19	10mins
2	2/01/19	12mins
3	3/01/19	18mins
4	4/01/19	13mins
5	7/01/19	16mins

Figure 4.13: Machine Ready Time

The above Figure 3.25 shows the time taken after a part is produced until the next part is loaded onto the machine. The total machine ready time for one day is 10mins.

The line graphs based on the data recorded during the First week of January are shown below:

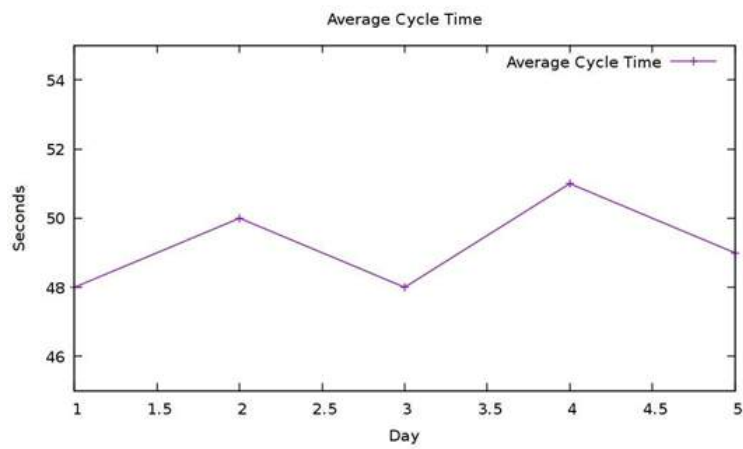


Figure 4.14: Average Cycle Time Graph

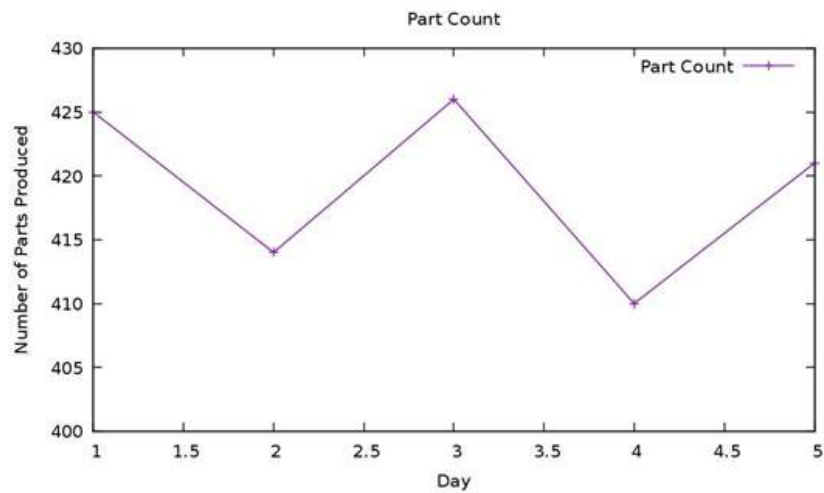


Figure 4.15: Part Count Graph

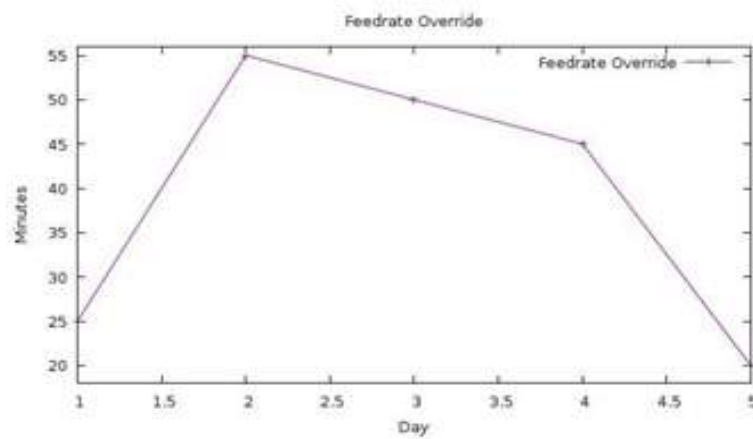


Figure 4.16: Feedrate Override Graph

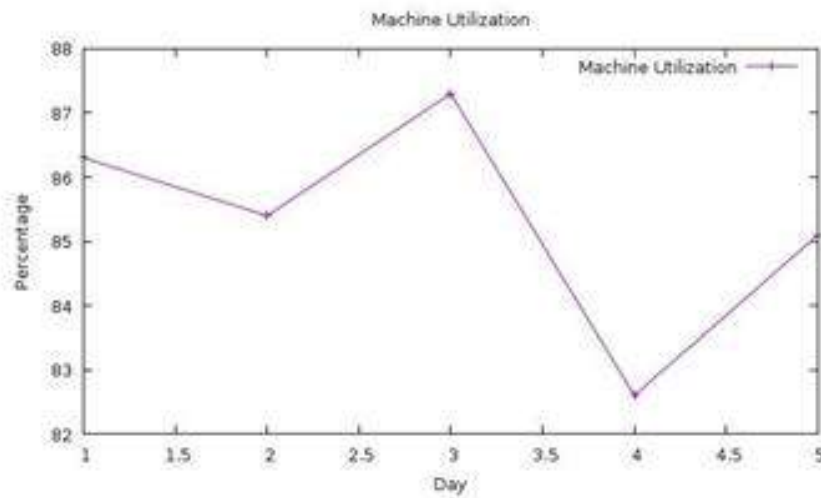


Figure 4.17: Utilization Graph

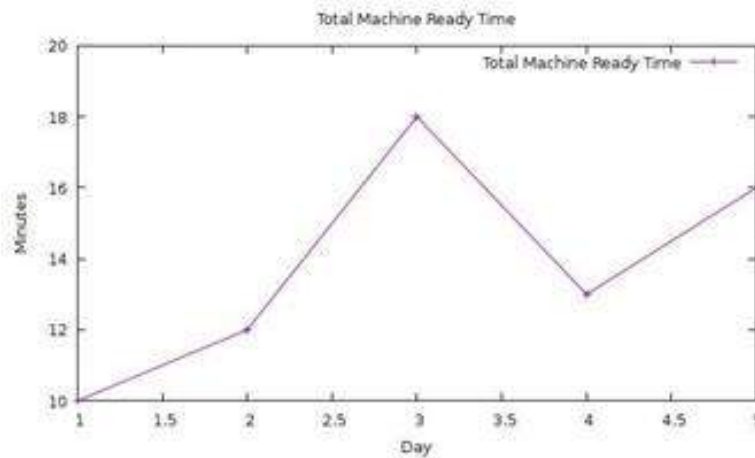


Figure 4.18: Machine Ready Time Graph

The complete comparison of these parameters can be easily established using these two bar graphs displayed below:

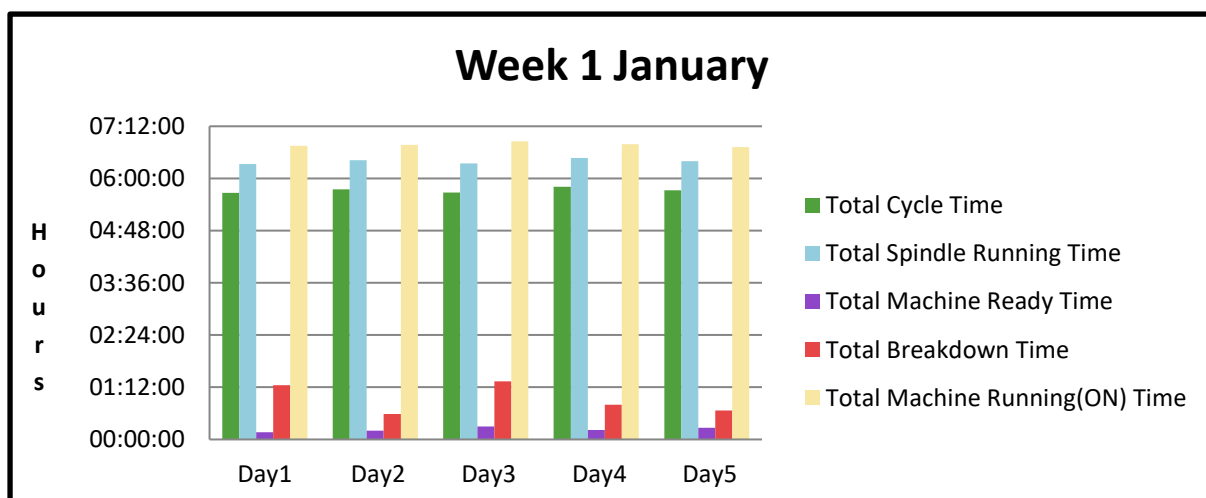


Figure 4.19: Comparison of Parameters

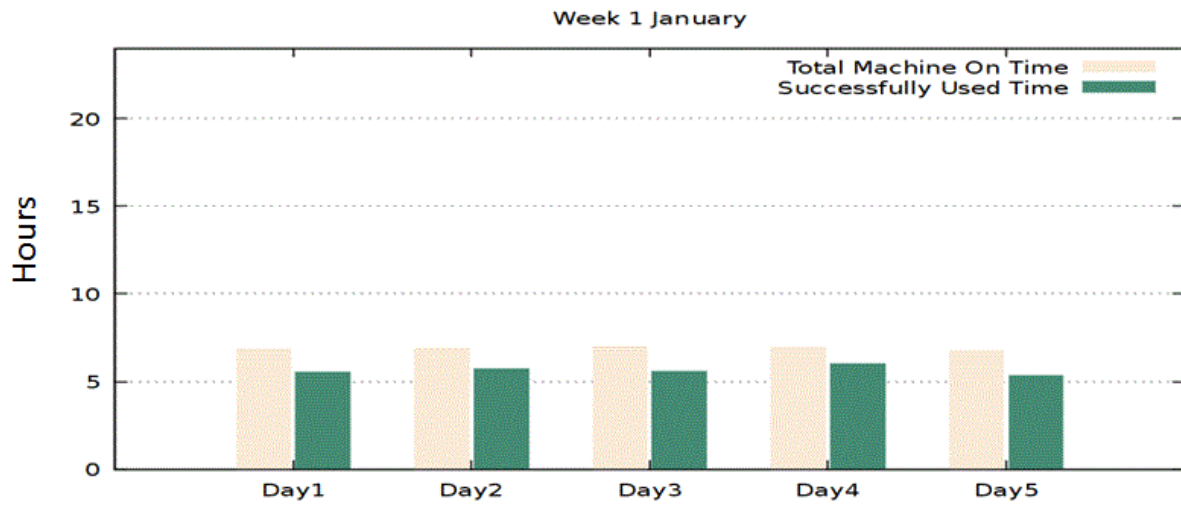


Figure 4.20: Utilization Bar Graph

On selecting each parameter for the second week of January, the corresponding table will be displayed. The display screens are shown below:

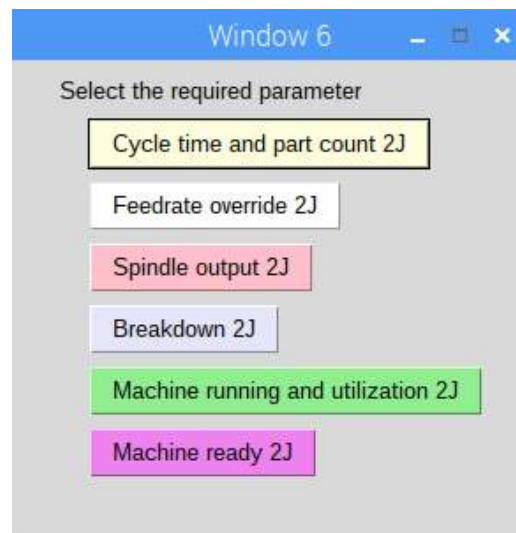


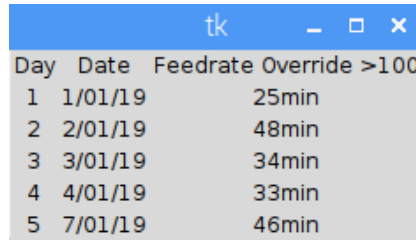
Figure 4.21: Parameters Week 2J

tk				
Day	Date	Average Cycle Time	Total Cycle Time	Part Count
1	1/01/19	49sec	5hr 41min 54sec	426
2	2/01/19	51sec	5hr 51min 54sec	414
3	3/01/19	51sec	5hr 52min 45sec	415
4	4/01/19	50sec	5hr 50min 50 sec	421
5	7/01/19	52sec	5hr 53min 36 sec	408

Figure 4.22: Cycle time and Part Count 2J

The above Figure 3.34 depicts that the average time for 1 part to be produced in a

day is 49 sec. The total number of parts produced in a day is 426 parts and the total time for 426 parts to be produced is 5hrs 41mins 54sec.



Day	Date	Feedrate Override >100
1	1/01/19	25min
2	2/01/19	48min
3	3/01/19	34min
4	4/01/19	33min
5	7/01/19	46min

Figure 4.23: Feedrate Override 2J

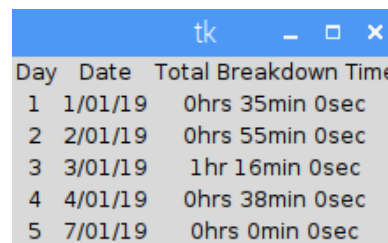
The feed rate override was greater than 100 for 25mins on 1st January 19. Figure 3.35 shows that the operator has increased the programmed feed rate over 100 for 25mins.



Day	Date	Total Spindle Running Hours
1	1/01/19	6hr 20min 54sec
2	2/01/19	6hr 21min 54sec
3	3/01/19	6hr 2min 45sec
4	4/01/19	6hr 10min 50 sec
5	7/01/19	6hr 23min 36 sec

Figure 4.24: Spindle Running Time 2J

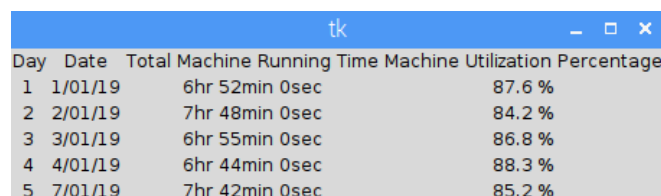
Figure 3.36 shows the amount of time the spindle has functioned for each day in a week. The total spindle running hours for 1 day is 5hrs 81mins 54sec.



Day	Date	Total Breakdown Time
1	1/01/19	0hrs 35min 0sec
2	2/01/19	0hrs 55min 0sec
3	3/01/19	1hr 16min 0sec
4	4/01/19	0hrs 38min 0sec
5	7/01/19	0hrs 0min 0sec

Figure 4.25: Breakdown Time 2J

Figure 3.37 depicts the amount of time in a day when the machine was unavailable or failed to perform its functions. The total breakdown time in a day is 35min.



Day	Date	Total Machine Running Time	Machine Utilization Percentage
1	1/01/19	6hr 52min 0sec	87.6 %
2	2/01/19	7hr 48min 0sec	84.2 %
3	3/01/19	6hr 55min 0sec	86.8 %
4	4/01/19	6hr 44min 0sec	88.3 %
5	7/01/19	7hr 42min 0sec	85.2 %

Figure 4.26: Machine Running and Utilization Percentage 2J

The total machine running time in a day was found to be 6hr 52min 0sec from Figure 3.38. The total machine utilization percentage for day 1 is 87.6%.

tk		
Day	Date	Total Machine Ready Time
1	1/01/19	10mins
2	2/01/19	11mins
3	3/01/19	12mins
4	4/01/19	10mins
5	7/01/19	15mins

Figure 4.27: Machine Ready Time 2J

The above Figure 3.39 shows the time taken after a part is produced until the next part is loaded onto the machine. The total machine ready time for one day is 10mins.

The line graphs based on the data recorded during the first week of January are shown below:

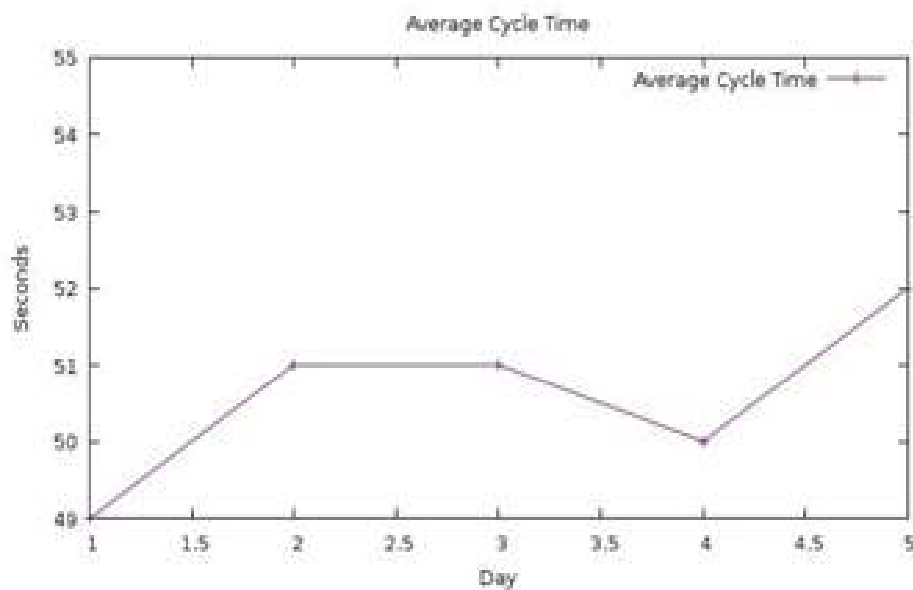


Figure 4.28: Average Cycle Time Graph 2J

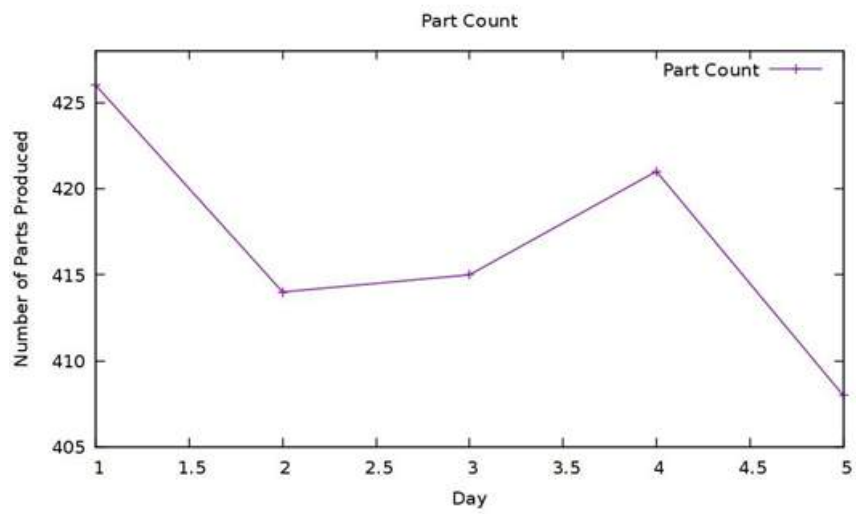


Figure 4.29: Part Count Graph 2J

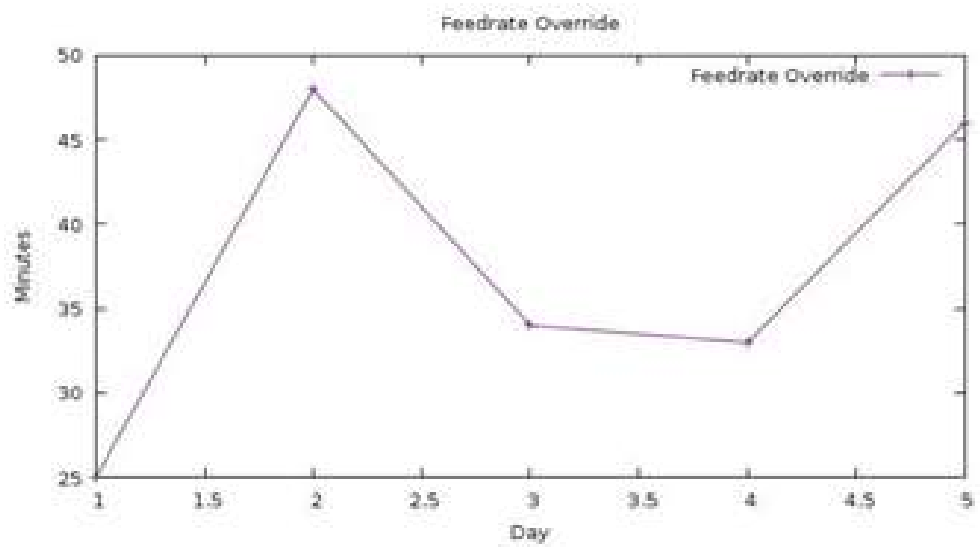


Figure 4.30: Feedrate Override Graph 2J

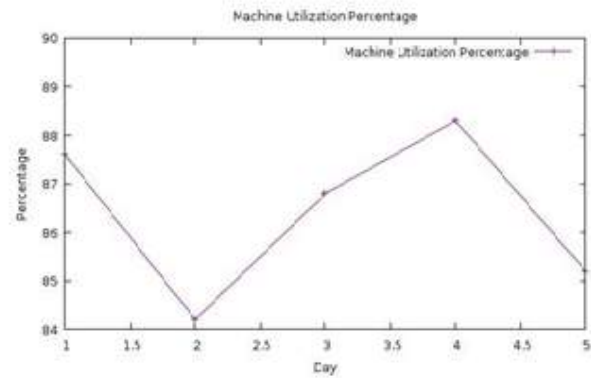


Figure 4.31: Utilization Graph 2J

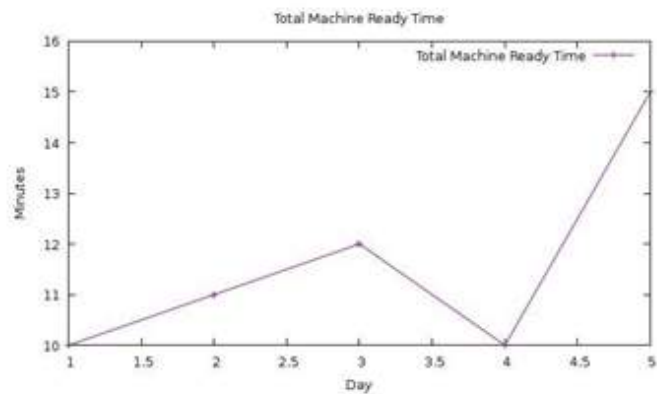


Figure 4.32: Machine Ready Time Graph

The complete comparison of these parameters can be easily established using these two bar graphs displayed below:

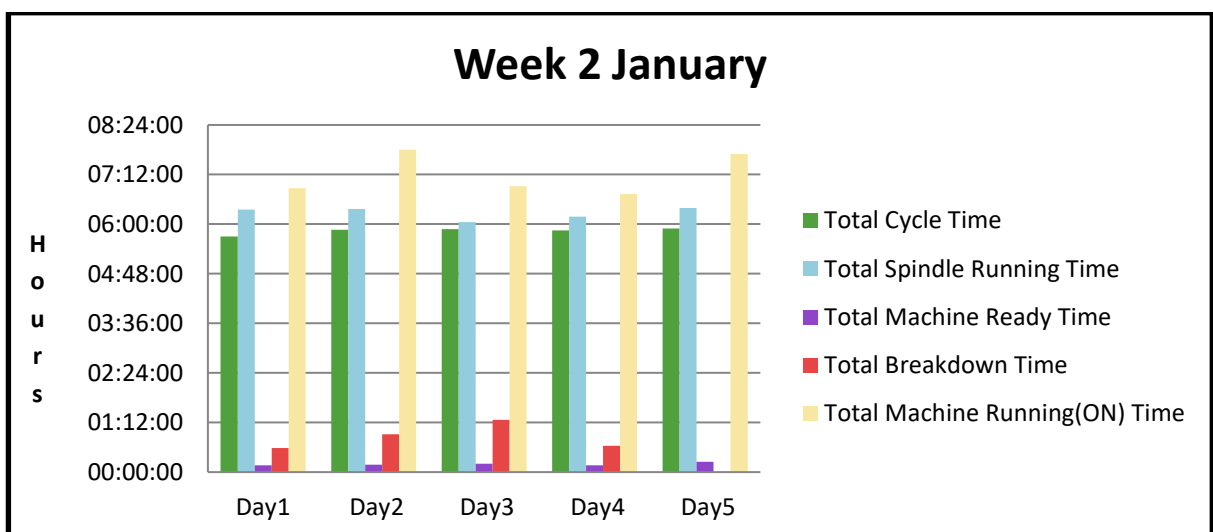


Figure 4.33: Comparison of Parameters Week 2

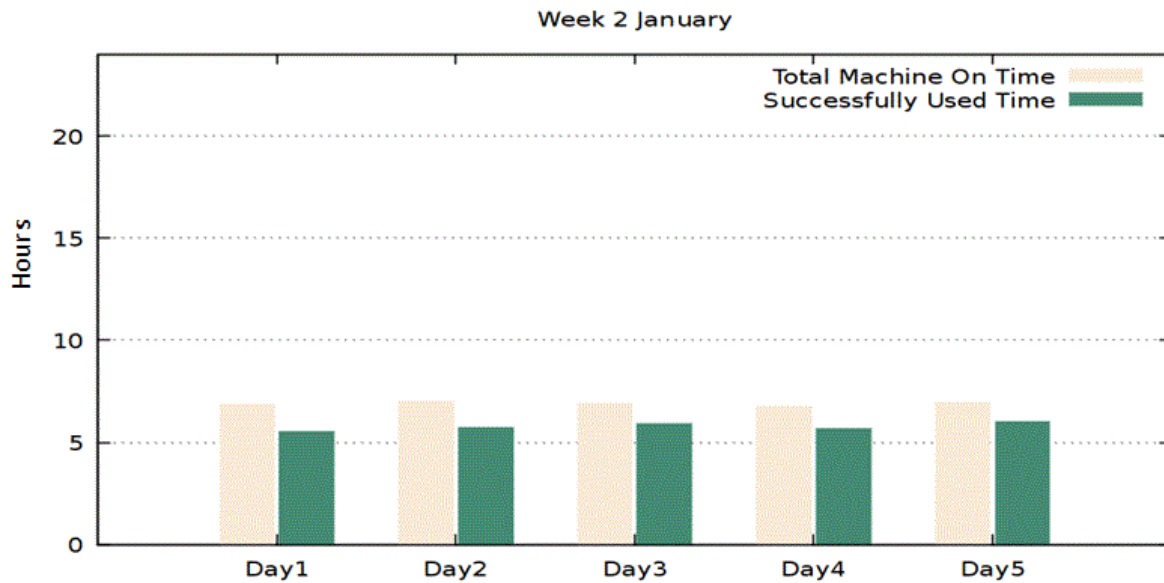


Figure 4.34: Utilization Bar Graph 2J

On analyzing the data, it is evident that breakdown time is significantly affecting the production. The Comparison of Breakdown Time with respect to Total Machine Run Time is displayed below.

Proper maintenance is very important for extending the life of the machine and increasing productivity. Despite this, for small manufacturing enterprises, there are not much tools or equipment available to understand the impact of the machine breakdown or production loss, and rarely are these variables measured. We intend to provide the correlation between the actual machine running time and the successfully utilized time so that we will know to what extent there are losses and this will help in better planning of maintenance and future activities. The tables and graphs help us in visualizing the impact of various parameters.

We aim to achieve maximum efficiency by improving utilization. One way to reduce breakdowns is to ensure proper alignment of all moving components and good lubrication and cooling systems. When a product is getting machined, if at that point of time the machine breaks down, there are chances that we may not be able to reuse that particular piece. In some cases, we may have to start with a fresh piece, which results in a loss. In this particular instance we were able to continue producing the part even after breakdown, but in other cases, it may be required to restart the process. Heat and contamination may also contribute to frequent breakdown. Minimization of

breakdown time can be achieved by scheduling proper maintenance. We should also make sure that the ready time should be nominal and minimize defective part count.

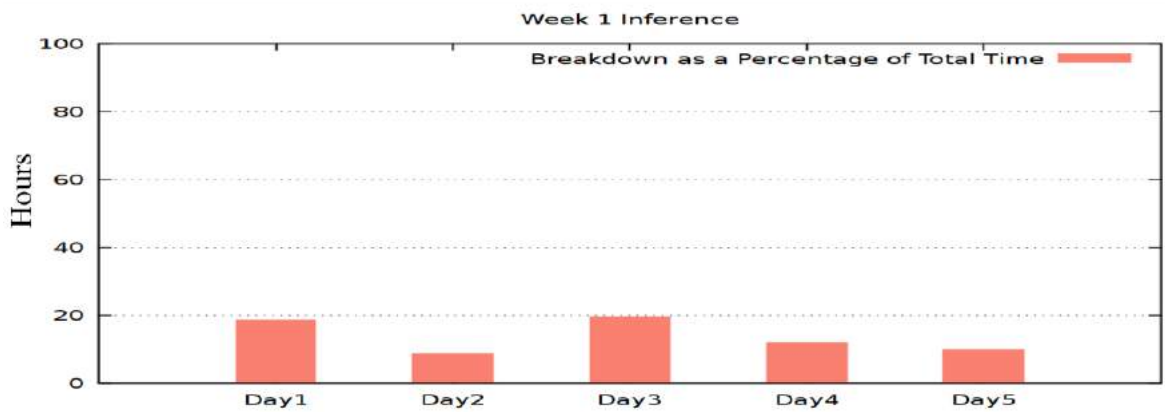


Figure 4.35: Comparison of Breakdown Time with respect to Total Machine Run Time

CHAPTER 5

CONCLUSION AND FUTURE ENHANCEMENT

Smart factories take the manufacturing industries a step ahead from traditional automation to a completely linked and adjustable system, which compels the companies to take up the latest industrial mechanisms. We have provided a feasible, cost effective solution using a raspberry pi to simulate an Industry 4.0 solution for CNC. This is a solution for small manufacturing companies to adopt new technologies for improving overall efficiency and become more competitive. We can capture the machine utilization parameters easily over a weekly period and simulate the acquired data in graphical form with the help of user interface screens. Data acquisition is done in real time so that the user can analyze the performance of the machine and the production rate at the current time. This method increases transparency, thereby giving insight on where scope is available to improve machine utilization. This helps the user to get more profit, production and higher efficiency. Adopting cost effective technology for monitoring and managing the utilization and efficiency of machine tools will help in reducing waste and becoming more productive.

Future enhancement can be done since the current model is for a single machine and this can be scaled up to connect to multiple machines. Additional parameters for monitoring can be incorporated. Root cause analysis can be performed to understand why the machine went into breakdown and hence reduce breakdown time. Protection from power fluctuations can be provided to the product. This product can be made into an app so that it can be used anywhere and anytime through mobile applications. A feature can be included in the app to allow control of the machine remotely.

APPENDIX A

PROGRAM CODES

A.1 Parameters

A.1.1 Machine Operating Mode: Auto/Manual

```
import RPi.GPIO as GPIO
import time
GPIO.setmode(GPIO.BCM)
GPIO.setup(18,GPIO.IN,pull_up_down=GPIO.PUD_UP)
GPIO.setup(17,GPIO.IN,pull_up_down=GPIO.PUD_UP)
while True:
    autoMode = GPIO.input(18)
    manualMode = GPIO.input(17)
    if autoMode == False:
        print("Automode is selected on CNC")
        time.sleep(0.2)
    elif manualMode == False:
        print("Manual mode is selected on CNC")
```

A.1.2 Part Program Running: Yes/No

```
import RPi.GPIO as GPIO
import time
GPIO.setmode(GPIO.BCM)
GPIO.setup(23,GPIO.IN,pull_up_down=GPIO.PUD_UP)
while True:
    partProgramSelected = GPIO.input(23)
    if partProgramSelected == False:
        print("Part program has been selected and is running")
```

A.1.3 Cycle Time and Part Count

```
import RPi.GPIO as GPIO
import time

GPIO.setmode(GPIO.BCM)
GPIO.setup(18, GPIO.IN, pull_up_down=GPIO.PUD_UP)
mem=0
partcount = 0
```



```

previouselapsedtime = 0
cycleelapsed = 0
elapsedtime = 0
currentCycleTime = 0
start= float(0)
stop= float(0)
while True:
    cyclestart = GPIO.input(18)
    time.sleep(0.3)
    if cyclestart == False and mem==0:
        print("start time recorded")
        start = time.time()
        mem=1
    if cyclestart == True and mem==1:
        print("stop time recorded")
        stop = time.time()
        mem=0

    currentCycleTime = stop-start
    cycleelapsed = cycleelapsed + (stop - start)
    if previouselapsedtime != cycleelapsed:
        partcount = partcount + 1
    averagecycletime = cycleelapsed/partcount
    previouselapsedtime = cycleelapsed
    print("current cycle time is", currentCycleTime)
    print("part count is", partcount)
    print("total cycle time elapsed is", cycleelapsed)
    print("average cycle time is", averagecycletime)

```

A.1.4 Feedrate Override

```

import RPi.GPIO as GPIO
import time

GPIO.setmode(GPIO.BCM)
GPIO.setup(17, GPIO.IN, pull_up_down=GPIO.PUD_UP)
ex=0
startnow = float(0)
stopnow = float(0)
while True:
    feedrateoverride = GPIO.input(17)
    if feedrateoverride == False and ex==0:
        startnow = time.time()

    print("feedrate override greater than 100 ALERT")
    ex=1
    if ex == 1 and feedrateoverride == True:
        stopnow = time.time()
    totalfeedrateoverridetime = (stopnow-startnow)

```

```

print("The total feedrate override time is", totalfeedrateoverridetime)
ex=0

```

A.1.5 Spindle Running Time

```

import RPi.GPIO as GPIO
import time
GPIO.setmode(GPIO.BCM)
GPIO.setup(22,GPIO.IN,pull_up_down=GPIO.PUD_UP)
var=0
hrs=0
rem=0
min=0
sec=0
minute=0
seconds=0
spindleRunningTime=0
totalSpindleRunningTime=0
start=float(0)
while True:
    spindleOutput = GPIO.input(22)
    time.sleep(0.3)
    def spindleFunction(totalSpindleRunningTime):
        if totalSpindleRunningTime>3600:
            hrs = totalSpindleRunningTime//3600
            rem = totalSpindleRunningTime%3600
            min = rem//60
            sec = rem%60
            print("Total Spindle Running Time Is",hrs,"hours",min,"minutes",sec,"seconds")
        elif totalSpindleRunningTime>60 and totalSpindleRunningTime<3600:
            minutes = totalSpindleRunningTime//60
            seconds = totalSpindleRunningTime%60
            print("Total Spindle Running Time Is",minutes,"minutes",seconds,"seconds")
        elif totalSpindleRunningTime<60:
            print("Total Spindle Running Time is",totalSpindleRunningTime,"seconds")
    if spindleOutput == False and var == 0:
        print("start time recorded")
        start = time.time()
        var = 1
    if spindleOutput == True and var ==1:
        print(" Stop Time Recorded")
        stop = time.time()
        var = 0
        spindleRunningTime = stop-start
        totalSpindleRunningTime = totalSpindleRunningTime+spindleRunningTime
        spindleFunction(totalSpindleRunningTime)

```

A.1.6 Machine Running Hours, Breakdown Time and Machine Utilization Percentage

```
import RPi.GPIO as GPIO
import time
GPIO.setmode(GPIO.BCM)
GPIO.setup(23,GPIO.IN,pull_up_down=GPIO.PUD_UP)
GPIO.setup(24,GPIO.IN,pull_up_down=GPIO.PUD_UP)
machineRunning = 0
totalMachineRunningTime = 0
temp = 0
machineUtilization = 0
cycleElapsed = 0
memr=0
breakdownTime =0
totalBreakdownTime =0
start = float(0)
stop = float (0)
while True:
    machineOutput = GPIO.input(23)
    alarmStatus = GPIO.input(24)
    time.sleep(0.3)
    def machineFunction(totalMachineRunningTime):
        if totalMachineRunningTime>3600:
            hrs = totalMachineRunningTime//3600
            rem = totalMachineRunningTime%3600
            min = rem//60
            sec = rem%60
            print("Total Machine Running Time Is",hrs,"hours",min,"minutes",sec,"seconds")
        elif totalMachineRunningTime>60 and totalMachineRunningTime<3600:
            minutes = totalMachineRunningTime//60
            seconds = totalMachineRunningTime%60
            print("Total Machine Running Time Is",minutes,"minutes",seconds,"seconds")
        elif totalMachineRunningTime<60:
            print("Total Machine Running Time is",totalMachineRunningTime,"seconds")
    if machineOutput == False and machineRunning == 0:
        print("Start time recorded")
        start = time.time()
        machineRunning = 1
    if machineOutput == True and machineRunning == 1:
        print("Stop time recorded")
        stop = time.time()
        machineRunning = 0
        machineRunningTime = stop-start
        totalMachineRunningTime = totalMachineRunningTime+machineRunningTime
        temp = (cycleElapsed*100)
        machineUtilization = temp//totalMachineRunningTime
        machineFunction(totalMachineRunningTime)
```

```

        print("Machine successfully used time",cycleElapsed)
        print("The CNC machine was utilized for",machineUtilization,"percent of the total time")
def alarmFunction(totalBreakdownTime):
    if totalBreakdownTime>3600:
        hrs = totalBreakdownTime//3600
        rem = totalBreakdownTime%3600
        min = rem//60
        sec = rem%60
        print("Total Breakdown Time is",hrs,"hours",min,"minutes",sec,"seconds")
    elif totalBreakdownTime>60 and totalBreakdownTime<3600:
        minutes = totalBreakdownTime//60
        seconds = totalBreakdownTime%60
        print("Total Breakdown Time is",minutes,"minutes",seconds,"seconds")
    elif totalBreakdownTime<60:
        print("Total Breakdown Time is",totalBreakdownTime,"seconds")
    if alarmStatus == False and memr==0:
        print("start time recorded")
        start = time.time()
        memr=1
    if cyclestart == True and memr==1:
        print("stop time recorded")
        stop = time.time()
        memr=0

    breakdownTime = stop-start
    totalBreakdownTime = breakdownTime + (stop - start)
    alarmFunction(totalBreakdownTime)

```

A.1.7 Machine Ready

```

import RPi.GPIO as GPIO
import time
GPIO.setmode(GPIO.BCM)
GPIO.setup(23,GPIO.IN,pull_up_down=GPIO.PUD_UP)
ready=0
readytime=0
totalReadyTime=0
start = float(0)
stop = float(0)
while True:
    machineReady = GPIO.input(23)

def readyFunction(totalReadyTime):
    if totalReadyTime>3600:
        hrs = totalReadyTime//3600
        rem = totalReadyTime%3600
        min = rem//60
        sec = rem%60

```

```

        print("Total Ready Time is",hrs,"hours",min,"minutes",sec,"seconds")
    elif totalReadyTime>60 and totalReadyTime<3600:
        minutes = totalReadyTime//60
        seconds = totalReadyTime%60
        print("Total Ready Time is",minutes,"minutes",seconds,"seconds")
    elif totalReadyTime<60:
        print("Total Ready Time is",totalReadyTime,"seconds")
if readyStatus == False and ready==0:
    print("start time recorded")
    start = time.time()
    ready=1
if cyclestart == True and ready==1:
    print("stop time recorded")
    stop = time.time()
    ready=0

    readytime = stop-start
    totalReadyTime = readyTime + (stop - start)
    readyFunction(total ReadyTime)

```

A.2 Program Code For GUI

```

import PySimpleGUI as sg

layout = [[sg.Text('Select the month')],
[sg.T(' '), sg.RealtimeButton(button_text=('January'),
button_color=('black', 'pink'))], [sg.T(' '),
sg.RealtimeButton(button_text=('February'), button_color=('black',
'light blue'))], [sg.T(' '), sg.RealtimeButton(button_text=('March'),
button_color=('black', 'violet'))], [sg.T(' '),
]

win1 = sg.Window('Window 1').Layout(layout)

win2_active =
False

win3_active =
False

win4_active =
False

win5_active =
False

win6_active =
False

win7_active =
False

win8_active =
False

win13_active =

```

```

False
win14_active =
False
win15_active =
False
win16_active =
False
win9_active =
False
win10_active =
False
win11_active =
False
win12_active =
False
win17_active =
False
win18_active =
False
win19_active =
False
win20_active =
False
win21_active =
False
win22_active =
False

while True:
    ev1, vals1 = win1.Read(timeout=1000000000)

    if not win2_active and ev1 ==
        'January': win2_active = True
        layout2 = [[sg.Text('Select the desired week')],
            [sg.T(' '), sg.RealtimeButton(button_text=('Week 1J'),
                button_color=('black', 'pink'))], [sg.T(' '),
            sg.RealtimeButton(button_text=('Week 2J'), button_color=('black',
                'pink'))], [sg.T(' '), sg.RealtimeButton(button_text=('Week 3J'),
                button_color=('black', 'pink'))], [sg.T(' '),
            sg.RealtimeButton(button_text=('Week 4J'), button_color=('black',
                'pink'))], [sg.T(' '),
            ]
        win2 = sg.Window('Window
        2').Layout(layout2)

        evel, valse1 =
        win2.Read(timeout=10000000000
        )

    if not win5_active and evel == 'Week
    1J': win5_active = True
    layout5 = [[sg.Text('Select the required parameter')],

```

```

[sg.T(' '), sg.RealtimeButton(button_text=('Cycle time and part count 1J'),
button_color=('black', 'light yellow'))], [sg.T(' '),
sg.RealtimeButton(button_text=('Feedrate override 1J'), button_color=('black', 'white'))],
[sg.T(' '), sg.RealtimeButton(button_text=('Spindle output 1J'),
button_color=('black', 'pink'))], [sg.T(' '),
sg.RealtimeButton(button_text=('Breakdown 1J'), button_color=('black',
'lavender'))],
[sg.T(' '), sg.RealtimeButton(button_text=('Machine running and utilization 1J'),
button_color=('black', 'light green'))], [sg.T(' '), sg.RealtimeButton(button_text=('Machine
ready 1J'), button_color=('black', 'violet'))],
]
win5 = sg.Window('Window
5').Layout(layout5) #start of table

evet1, valset1 = win5.Read(timeout=1000000000)

if not win17_active and evet1 == 'Cycle time and part
count 1J': win17_active = True
from tkinter
import *
window=Tk()
l1=Label(window,
text="Day")
l1.grid(row=0,column=
0) l1=Label(window,
text="Date")
l1.grid(row=0,column=
1)
l1=Label(window, text="Average
Cycle Time")
l1.grid(row=0,column=2)
l1=Label(window, text="Total
Cycle Time")
l1.grid(row=0,column=3)
l1=Label(window,
text="Part Count")
l1.grid(row=0,column=4)
l1=Label(window, text="1")
l1.grid(row=1,column=0)
l1=Label(window, text="2")
l1.grid(row=2,column=0)
l1=Label(window, text="3")
l1.grid(row=3,column=0)
l1=Label(window, text="4")
l1.grid(row=4,column=0)
l1=Label(window, text="5")
l1.grid(row=5,column=0)
l1=Label(window,
text="1/01/19")
l1.grid(row=1,column=1)
l1=Label(window,

```

```

text="48sec")
l1.grid(row=1,column=2)
l1=Label(window, text="5hr
40min 0sec")
l1.grid(row=1,column=3)
l1=Label(window, text="425")
l1.grid(row=1,column=4)
l1=Label(window,
text="2/01/19")
l1.grid(row=2,column=1)
l1=Label(window,
text="50sec")
l1.grid(row=2,column=2)
l1=Label(window, text="5hr
45min 0sec")
l1.grid(row=2,column=3)
l1=Label(window, text="414")
l1.grid(row=2,column=4)
l1=Label(window,
text="3/01/19")
l1.grid(row=3,column=1)
l1=Label(window,
text="48sec")
l1.grid(row=3,column=2)
l1=Label(window, text="5hr 40min
48sec 0sec")
l1.grid(row=3,column=3)
l1=Label(window,
text="426")
l1.grid(row=3,column=4)
l1=Label(window,
text="4/01/19")
l1.grid(row=4,column=1)
l1=Label(window,
text="51sec")
l1.grid(row=4,column=2)
l1=Label(window, text="5hr
48min 30sec")
l1.grid(row=4,column=3)
l1=Label(window,
text="410")
l1.grid(row=4,column=4)
l1=Label(window,
text="7/01/19")
l1.grid(row=5,column=1)
l1=Label(window,
text="49sec")
l1.grid(row=5,column=2)
l1=Label(window, text="5hr
43min 49sec")
l1.grid(row=5,column=3)

```



```

l1=Label(window,
text="421")
l1.grid(row=5,column=4)
window.mainloop()

win17 = sg.Window('Window 17')

if win17_active:
    evet1, valset1 = win17.Read(timeout=1000000000)

evet2, valset2 = win5.Read(timeout=10)
if not win18_active and evet2 == 'Feedrate
    override 1J': win18_active = True
from tkinter
import *
window=Tk()
l1=Label(window,
text="Day")
l1.grid(row=0,column=0)
l1=Label(window,
text="Date")

l1.grid(row=0,column=1)
l1=Label(window, text="Feedrate
Override >100")
l1.grid(row=0,column=2)
l1=Label(window, text="1")
l1.grid(row=1,column=0)
l1=Label(window, text="2")
l1.grid(row=2,column=0)
l1=Label(window, text="3")
l1.grid(row=3,column=0)
l1=Label(window, text="4")
l1.grid(row=4,column=0)
l1=Label(window, text="5")
l1.grid(row=5,column=0)
l1=Label(window,
text="1/01/19")
l1.grid(row=1,column=1)
l1=Label(window,
text="25min")
l1.grid(row=1,column=2)
l1=Label(window,
text="2/01/19")
l1.grid(row=2,column=1)
l1=Label(window,
text="55min")
l1.grid(row=2,column=2)
l1=Label(window,
text="3/01/19")
l1.grid(row=3,column=1)
l1=Label(window,

```

```

text="50min")
l1.grid(row=3,column=2)
l1=Label(window,
text="4/01/19")
l1.grid(row=4,column=1)
l1=Label(window,
text="45min")
l1.grid(row=4,column=2)
l1=Label(window,
text="7/01/19")
l1.grid(row=5,column=1)
l1=Label(window,
text="20min")
l1.grid(row=5,column=2)
window.mainloop()

```

```

win18 = sg.Window('Window 18')

```

```

if win18_active:
    evet2, valset2
    =win18.Read(time
    out=1000000000)
    evet3, valset3 =
    win5.Read(timeou
    t=1000000000)

if not win19_active and evet3 ==
'Spindle output 1J': win19_active =
True
from tkinter
import *
window=Tk()
l1=Label(window,
text="Day")
l1.grid(row=0,column=
0) l1=Label(window,
text="Date")
l1.grid(row=0,column=
1)
l1=Label(window, text="Total Spindle
Running Hours") l1.grid(row=0,column=2)
l1=Label(window, text="1")
l1.grid(row=1,column=0)
l1=Label(window, text="2")
l1.grid(row=2,column=0)
l1=Label(window, text="3")
l1.grid(row=3,column=0)
l1=Label(window, text="4")
l1.grid(row=4,column=0)
l1=Label(window, text="5")
l1.grid(row=5,column=0)

```

```

l1=Label(window,
text="1/01/19")
l1.grid(row=1,column=1)
l1=Label(window, text="5hr
80min 0sec")
l1.grid(row=1,column=2)

l1=Label(window,
text="2/01/19")
l1.grid(row=2,column=1)
l1=Label(window, text="5hr
85min 0sec")
l1.grid(row=2,column=2)
l1=Label(window,
text="3/01/19")
l1.grid(row=3,column=1)
l1=Label(window, text="5hr
80min 48sec")
l1.grid(row=3,column=2)
l1=Label(window,
text="4/01/19")
l1.grid(row=4,column=1)
l1=Label(window, text="5hr
88min 30sec")
l1.grid(row=4,column=2)


l1=Label(window,
text="7/01/19")
l1.grid(row=5,column=1)
l1=Label(window, text="5hr
83min 49sec")
l1.grid(row=5,column=2)


window.mainloop()

win19 =
sg.Window('Window 19') if
win19_active:
    evet3, valset3 =
win19.Read(timeout=1000000000) evet4,
valset4 = win5.Read(timeout=1000000000)
if not win20_active and evet4 ==
    'Breakdown 1J': win20_active = True
from tkinter
import *
window=Tk()
l1=Label(window,
text="Day")
l1.grid(row=0,column=
0) l1=Label(window,
text="Date")
l1.grid(row=0,column=
1)

```

```

l1=Label(window,
text="Breakdown Hours")
l1.grid(row=0,column=2)
l1=Label(window, text="1")
l1.grid(row=1,column=0)
l1=Label(window, text="2")

l1.grid(row=2,column=0)
l1=Label(window, text="3")
l1.grid(row=3,column=0)
l1=Label(window, text="4")
l1.grid(row=4,column=0)
l1=Label(window, text="5")
l1.grid(row=5,column=0)
l1=Label(window,
text="1/01/19")
l1.grid(row=1,column=1)
l1=Label(window, text="1hr
15min 0sec")
l1.grid(row=1,column=2)

l1=Label(window,
text="2/01/19")
l1.grid(row=2,column=1)
l1=Label(window, text="0hr
35min 0sec")
l1.grid(row=2,column=2)
l1=Label(window,
text="3/01/19")
l1.grid(row=3,column=1)
l1=Label(window, text="1hr
20min 0sec")
l1.grid(row=3,column=2)

l1=Label(window,
text="4/01/19")
l1.grid(row=4,column=1)
l1=Label(window, text="0hr
48min 0sec")
l1.grid(row=4,column=2)

l1=Label(window,
text="7/01/19")
l1.grid(row=5,column=1)
l1=Label(window, text="0hr
40min 0sec")
l1.grid(row=5,column=2)

window.mainloop()
win20 = sg.Window('Window 20')

if win20_active:

```

```

    evet4, valset4 =
win20.Read(timeout=1000000000) evet5,
valset5 = win5.Read(timeout=1000000000)
if not win21_active and evet5 == 'Machine running and
utilization 1J': win21_active = True
from tkinter
import *
window=Tk()
l1=Label(window,
text="Day")
l1.grid(row=0,column=
0) l1=Label(window,
text="Date")
l1.grid(row=0,column=
1)
l1=Label(window, text="Total Machine
Running Time") l1.grid(row=0,column=2)
l1=Label(window, text="Machine
Utilization Percentage")
l1.grid(row=0,column=3)
l1=Label(window,
text="1")
l1.grid(row=1,column=0)
l1=Label(window,
text="2")
l1.grid(row=2,column=0)
l1=Label(window,
text="3")
l1.grid(row=3,column=0)
l1=Label(window,
text="4")
l1.grid(row=4,column=0)
l1=Label(window,
text="5")
l1.grid(row=5,column=0)
l1=Label(window,
text="1/01/19")
l1.grid(row=1,column=1)
l1=Label(window, text="6hr
45min 10sec")
l1.grid(row=1,column=2)
l1=Label(window,
text="86.3 %")
l1.grid(row=1,column=3)
l1=Label(window,
text="2/01/19")

```

```

l1.grid(row=2,column=1)
l1=Label(window, text="6hr
46min 18sec")
l1.grid(row=2,column=2)
l1=Label(window,
text="85.4 %")
l1.grid(row=2,column=3)
l1=Label(window,
text="3/01/19")
l1.grid(row=3,column=1)
l1=Label(window, text="6hr
51min 23sec")
l1.grid(row=3,column=2)
l1=Label(window,
text="87.3 %")
l1.grid(row=3,column=3)
l1=Label(window,
text="4/01/19")
l1.grid(row=4,column=1)
l1=Label(window, text="6hr
47min 28sec")
l1.grid(row=4,column=2)
l1=Label(window,
text="82.6 %")
l1.grid(row=4,column=3)
l1=Label(window,
text="7/01/19")
l1.grid(row=5,column=1)
l1=Label(window, text="6hr
43min 16sec")
l1.grid(row=5,column=2)
l1=Label(window, text="85.1
%") l1.grid(row=5,column=3)
window.mainloop()
win21 = sg.Window('Window 21')

if win21_active:
    evet5, valset5 = win21.Read(timeout=1000000000)
    evet6, valset6 = win5.Read(timeout=1000000000)

if not win22_active and evet6 ==
'Machine ready 1J': win22_active =
True
from tkinter
import *
window=Tk()
l1=Label(window,
text="Day")
l1.grid(row=0,column=
0) l1=Label(window,
text="Date")
l1.grid(row=0,column=

```

```

1)
l1=Label(window, text="Total Machine
Running Time") l1.grid(row=0,column=2)
l1=Label(window, text="Machine
Utilization Percentage")
l1.grid(row=0,column=3)
l1.grid(row=0,column=2)
l1=Label(window,
text="1")
l1.grid(row=1,column=0)
l1=Label(window,
text="2")
l1.grid(row=2,column=0)
l1=Label(window,
text="3")
l1.grid(row=3,column=0)
l1=Label(window,
text="4")
l1.grid(row=4,column=0)
l1=Label(window,
text="5")
l1.grid(row=5,column=0)
l1=Label(window,
text="1/01/19")
l1.grid(row=1,column=1)
l1=Label(window,
text="10mins")
l1.grid(row=1,column=2)

l1=Label(window,
text="2/01/19")
l1.grid(row=2,column=1)
l1=Label(window,
text="12mins")
l1.grid(row=2,column=2)
l1=Label(window,text="3
/01/19")
l1.grid(row=3,column=1)
l1=Label(window,
text="18mins")
l1.grid(row=3,column=2)

l1=Label(window,
text="4/01/19")
l1.grid(row=4,column=1)
l1=Label(window,
text="13mins")
l1.grid(row=4,column=2)

l1=Label(window,
text="7/01/19")
l1.grid(row=5,column=1)

```

```

        l1=Label(window,
        text="16mins")
        l1.grid(row=5,column=2)

        window.mainloop()
        win22 =
        sg.Window('Window 22') if
        win22_active:
            evet6, valset6 =
            win22.Read(timeout=1000000000) #stop
            table 1

        if win5_active:
            eve1, valse1 = win5.Read(timeout=10000000)

        eve2, valse2 =
        win2.Read(timeout=1000000) if
        not win6_active and eve2 ==
        'Week 2J':
            win6_active = True
            layout6 = [[sg.Text('Select the required parameter')],
            [sg.T(' '), sg.RealtimeButton(button_text=('Cycle time and part count 2J'),
            button_color=('black', 'light yellow'))], [sg.T(' '),
            sg.RealtimeButton(button_text=('Feedrate override 2J'), button_color=('black', 'white'))],
            [sg.T(' '), sg.RealtimeButton(button_text=('Spindle output 2J'), button_color=('black',
            'pink'))],
            [sg.T(' '), sg.RealtimeButton(button_text=('Breakdown 2J'), button_color=('black',
            'lavender'))],
            [sg.T(' '), sg.RealtimeButton(button_text=('Machine running and utilization 2J'),
            button_color=('black', 'light green'))], [sg.T(' '), sg.RealtimeButton(button_text=('Machine
            ready 2J'), button_color=('black', 'violet'))],
            [sg.T('')],

            ]

        win6 = sg.Window('Window
        6').Layout(layout6) #strt table 2

        if win6_active:
            eve2, valse2 = win6.Read(timeout=10000000)

        eve3, valse3 =
        win2.Read(timeout=10000) if
        not win7_active and eve3 ==
        'Week 3J':
            win7_active = True
            layout7 = [[sg.Text('Select the required parameter')],
            [sg.T(' '), sg.RealtimeButton(button_text=('Cycle time and part count 3J'),
            button_color=('black', 'light yellow'))], [sg.T(' '),
            sg.RealtimeButton(button_text=('Feedrate override 3J'), button_color=('black', 'white'))],
            [sg.T(' '), sg.RealtimeButton(button_text=('Spindle output 3J'),
            button_color=('black', 'pink'))], [sg.T(' '),
            sg.RealtimeButton(button_text=('Breakdown 3J'), button_color=('black',

```



```

        'lavender'))],
        [sg.T(' '), sg.RealtimeButton(button_text=('Machine running and utilization 3J'),
        button_color=('black', 'light green'))], [sg.T(' '), sg.RealtimeButton(button_text=('Machine
        ready 3J'), button_color=('black', 'violet'))],
        [sg.T('')],

        ]

        win7 = sg.Window('Window 7').Layout(layout7)

        if win7_active:
            eve3, valse3 = win7.Read(timeout=10000000)

        eve4, valse4 = win2.Read(timeout=10000)

        if not win8_active and eve4 ==
            'Week 4J': win8_active = True
            layout8 = [[sg.Text('Select the required parameter')],
            [sg.T(' '), sg.RealtimeButton(button_text=('Cycle time and part count 4J'),
            button_color=('black', 'light yellow'))], [sg.T(' '),
            sg.RealtimeButton(button_text=('Feedrate override 4J'), button_color=('black', 'white'))],
            [sg.T(' '), sg.RealtimeButton(button_text=('Spindle output 4J'),
            button_color=('black', 'pink'))], [sg.T(' '),
            sg.RealtimeButton(button_text=('Breakdown 4J'), button_color=('black',
            'lavender'))],
            [sg.T(' '), sg.RealtimeButton(button_text=('Machine running and utilization 4J'),
            button_color=('black', 'light green'))], [sg.T(' '), sg.RealtimeButton(button_text=('Machine
            ready 4J'), button_color=('black', 'violet'))],

            [sg.T('')],

            ]

            win8 = sg.Window('Window 8').Layout(layout8)

            if win8_active:
                eve4, valse4 =
                win8.Read(timeout=10000000) if
                win2_active:
                    ev2, vals2 =
                win2.Read(timeout=100) ev3,
                vals3 =
                win1.Read(timeout=100000)

            if not win4_active and ev3 ==
                'March': win4_active =
                True
                layout4 = [[sg.Text('Select the week')],
                [sg.T(' '), sg.RealtimeButton(button_text=('Week 1M'),

```

```

button_color=('black', 'violet'))], [sg.T(' '),
sg.RealtimeButton(button_text=('Week 2M'), button_color=('black',
'violet'))],

[sg.T(' '), sg.RealtimeButton(button_text=('Week 3M'),
button_color=('black', 'violet'))], [sg.T(' '),
sg.RealtimeButton(button_text=('Week 4M'), button_color=('black',
'violet'))], [sg.T('')],

]

win4 = sg.Window('Window 4').Layout(layout4)

#strt window 13 to 16
eval, valsa1 = win4.Read(timeout=10000)
if not win13_active and eval ==
    'Week 1M': win13_active = True
    layout13 = [[sg.Text('Select the required parameter')],
[sg.T(' '), sg.RealtimeButton(button_text=('Cycle time and part count 1M'),
button_color=('black', 'light yellow'))], [sg.T(' '),
sg.RealtimeButton(button_text=('Feedrate override 1M'), button_color=('black', 'white'))],
[sg.T(' '), sg.RealtimeButton(button_text=('Spindle output 1M'),
button_color=('black', 'pink'))], [sg.T(' '),
sg.RealtimeButton(button_text=('Breakdown 1M'), button_color=('black',
'lavender'))],
[sg.T(' '), sg.RealtimeButton(button_text=('Machine running and utilization 1M'),
button_color=('black', 'light green'))], [sg.T(' '), sg.RealtimeButton(button_text=('Machine
ready 1M'), button_color=('black', 'violet'))],

[sg.T('')],

]

win13 = sg.Window('Window 13').Layout(layout13)

if win13_active:
    eval, valsa1 = win13.Read(timeout=10000000)

#start14
eva2, valsa2 = win4.Read(timeout=10000)
if not win14_active and eva2 ==
    'Week 2M': win14_active = True
    layout14 = [[sg.Text('Select the required parameter')],
[sg.T(' '), sg.RealtimeButton(button_text=('Cycle time and part count 2M'),
button_color=('black', 'light yellow'))], [sg.T(' '),
sg.RealtimeButton(button_text=('Feedrate override 2M'), button_color=('black', 'white'))],
[sg.T(' '), sg.RealtimeButton(button_text=('Spindle output 2M'),
button_color=('black', 'pink'))], [sg.T(' '),
sg.RealtimeButton(button_text=('Breakdown 2M'), button_color=('black',
'lavender'))],
[sg.T(' '), sg.RealtimeButton(button_text=('Machine running and utilization 2M'),
button_color=('black', 'light green'))], [sg.T(' '), sg.RealtimeButton(button_text=('Machine
ready 2M'), button_color=('black', 'violet'))],

[sg.T('')],

```

```

]
win14 = sg.Window('Window 14').Layout(layout14)

if win14_active:
    eva2, valsa2 = win14.Read(timeout=10000000)

#start15
eva3, valsa3 = win4.Read(timeout=10000)
if not win15_active and eva3 ==
    'Week 3M': win15_active = True
    layout15 = [[sg.Text('Select the required parameter')],
    [sg.T(' '), sg.RealtimeButton(button_text=('Cycle time and part count 3M'),
    button_color=('black', 'light yellow'))], [sg.T(' '),
    sg.RealtimeButton(button_text=('Feedrate override 3M'), button_color=('black', 'white'))],
    [sg.T(' '), sg.RealtimeButton(button_text=('Spindle output 3M'),
    button_color=('black', 'pink'))], [sg.T(' '),
    sg.RealtimeButton(button_text=('Breakdown 3M'), button_color=('black',
    'lavender'))],
    [sg.T(' '), sg.RealtimeButton(button_text=('Machine running and utilization 3M'),
    button_color=('black', 'light green'))], [sg.T(' '), sg.RealtimeButton(button_text=('Machine
    ready 3M'), button_color=('black', 'violet'))],
    [sg.T('')],
    ]

    win15 = sg.Window('Window 15').Layout(layout15)

if win15_active:
    eva3, valsa3 = win15.Read(timeout=10000000)

#start16
eva4, valsa4 = win4.Read(timeout=10000)

if not win16_active and eva4 ==
    'Week 4M': win16_active = True
    layout16 = [[sg.Text('Select the required parameter')],
    [sg.T(' '), sg.RealtimeButton(button_text=('Cycle time and part count 4M'),
    button_color=('black', 'light yellow'))], [sg.T(' '),
    sg.RealtimeButton(button_text=('Feedrate override 4M'), button_color=('black', 'white'))],
    [sg.T(' '), sg.RealtimeButton(button_text=('Spindle output 4M'),
    button_color=('black', 'pink'))], [sg.T(' '),
    sg.RealtimeButton(button_text=('Breakdown 4M'), button_color=('black',
    'lavender'))],
    [sg.T(' '), sg.RealtimeButton(button_text=('Machine running and utilization 4M'),
    button_color=('black', 'light green'))], [sg.T(' '), sg.RealtimeButton(button_text=('Machine
    ready 4M'), button_color=('black', 'violet'))],
    [sg.T('')],
    ]

    win16 = sg.Window('Window 16').Layout(layout16)

```

```

if win16_active:
    eva4, valsa4 = win16.Read(timeout=10000000)

#stop
if win4_active:
    ev4, vals4 =
    win4.Read(timeout=100) if
    ev4 is None or ev4 ==
    'Exit':
        win4_active =
        False win4.Close()

ev2, vals2 = win1.Read(timeout=100000)

if not win3_active and ev2 ==
    'February': win3_active = True
    layout3 = [[sg.Text('Select the week')],
    [sg.T(' '), sg.RealtimeButton(button_text=('Week 1F'),
    button_color=('black', 'light blue'))], [sg.T(' '),
    sg.RealtimeButton(button_text=('Week 2F'), button_color=('black',
    'light blue'))], [sg.T(' '), sg.RealtimeButton(button_text=('Week
    3F'), button_color=('black', 'light blue'))], [sg.T(' '),
    sg.RealtimeButton(button_text=('Week 4F'), button_color=('black',
    'light blue'))], [sg.T('')],
    ]

    win3 = sg.Window('Window 3').Layout(layout3)

#start of window 9 till 12
evb1, valsb1 = win3.Read(timeout=10000)

if not win9_active and evb1 ==
    'Week 1F': win9_active = True
    layout9 = [[sg.Text('Select the required parameter')],
    [sg.T(' '), sg.RealtimeButton(button_text=('Cycle time and part count 1F'),
    button_color=('black', 'light yellow'))], [sg.T(' '),
    sg.RealtimeButton(button_text=('Feedrate override 1F'), button_color=('black', 'white'))],
    [sg.T(' '), sg.RealtimeButton(button_text=('Spindle output 1F'),
    button_color=('black', 'pink'))], [sg.T(' '),
    sg.RealtimeButton(button_text=('Breakdown 1F'), button_color=('black',
    'lavender'))],
    [sg.T(' '), sg.RealtimeButton(button_text=('Machine running and utilization 1F'),
    button_color=('black', 'light green'))], [sg.T(' '), sg.RealtimeButton(button_text=('Machine
    ready 1F'), button_color=('black', 'violet'))],
    [sg.T('')],
    ]

```

```

win9 = sg.Window('Window 9').Layout(layout9)

if win9_active:
    evb1, valsb1 =
win9.Read(timeout=10000000) if
    evb1 is None or evb1 == 'Exit':
        win9_active =
            False
        win9.Close()
#start10
evb2, valsb2 = win3.Read(timeout=10000)

if not win10_active and evb2 ==
    'Week 2F': win10_active = True
    layout10 = [[sg.Text('Select the required parameter')],
[sg.T(' '), sg.RealtimeButton(button_text=('Cycle time and part count 2F'),
    button_color=('black', 'light yellow'))], [sg.T(' '),
    sg.RealtimeButton(button_text=('Feedrate override 2F'), button_color=('black', 'white'))],
[sg.T(' '), sg.RealtimeButton(button_text=('Spindle output 2F'),
    button_color=('black', 'pink'))], [sg.T(' '),
    sg.RealtimeButton(button_text=('Breakdown 2F'), button_color=('black',
    'lavender'))],
[sg.T(' '), sg.RealtimeButton(button_text=('Machine running and utilization 2F'),
    button_color=('black', 'light green'))], [sg.T(' '), sg.RealtimeButton(button_text=('Machine
    ready 2F'), button_color=('black', 'violet'))],
[sg.T('')],
    ]

win10 = sg.Window('Window 10').Layout(layout10)

if win10_active:
    evb2, valsb2 =
win10.Read(timeout=10000000) if
    evb2 is None or evb2 == 'Exit':
        win10_active = False
        win10.Close()
#start11
evb3, valsb3 = win3.Read(timeout=10000)

if not win11_active and evb3 ==
    'Week 3F': win11_active = True
    layout11 = [[sg.Text('Select the required parameter')],
[sg.T(' '), sg.RealtimeButton(button_text=('Cycle time and part count 3F'),
    button_color=('black', 'light yellow'))], [sg.T(' '),
    sg.RealtimeButton(button_text=('Feedrate override 3F'), button_color=('black', 'white'))],
[sg.T(' '), sg.RealtimeButton(button_text=('Spindle output 3F'),

```

```

        button_color=('black', 'pink'))], [sg.T(' '),
        sg.RealtimeButton(button_text=('Breakdown 3F'), button_color=('black',
        'lavender'))],
        [sg.T(' '), sg.RealtimeButton(button_text=('Machine running and utilization 3F'),
        button_color=('black', 'light green'))], [sg.T(' '), sg.RealtimeButton(button_text=('Machine ready
        3F'), button_color=('black', 'violet'))],
        [sg.T('')],
        ]

win11 = sg.Window('Window 11').Layout(layout11)

if win11_active: evb3,
    valsb3 =
        win11.Read(timeout=10000000) if
        evb3 is None or evb3 == 'Exit':
            win11_active = False win11.Close()
#start12
evb4, valsb4 = win3.Read(timeout=10000)

if not win12_active and evb4 ==
    'Week 4F': win12_active = True
    layout12 = [[sg.Text('Select the required parameter')],
    [sg.T(' '), sg.RealtimeButton(button_text=('Cycle time and part count 4F'), button_color=('black',
    'light yellow'))], [sg.T(' '), sg.RealtimeButton(button_text=('Feedrate override 4F'),
    button_color=('black', 'white'))], [sg.T(' '), sg.RealtimeButton(button_text=('Spindle output 4F'),
    button_color=('black', 'pink'))], [sg.T(' '),
    sg.RealtimeButton(button_text=('Breakdown 4F'), button_color=('black',
    'lavender'))],
    [sg.T(' '), sg.RealtimeButton(button_text=('Machine running and utilization 4F'),
    button_color=('black', 'light green'))], [sg.T(' '), sg.RealtimeButton(button_text=('Machine ready
    4F'), button_color=('black', 'violet'))],
    [sg.T('')],
    ]

win12 = sg.Window('Window 12').Layout(layout12)

if win12_active: evb4,
    valsb4 =
        win12.Read(timeout=10000000) if
        evb4 is None or evb4 == 'Exit':
            win12_active = False win12.Close()
#stop
if win3_active: ev3,
    valsb3 =
        win3.Read(timeout=100) if ev3 is
        None or ev3 == 'Exit':
            win3_active = False
            win3.Close()

```

A.3 Program Code For Parameter Comparison Bar Graph

```

red = "#cd3333"; green = "#458b74"; blue = "#87cefa"; purple =
"#cd96cd"; peach = "#FAEBD7"; set ydata time
set timefmt "%H:%M:%S"
set
format y "%H:%M:%S" set

```

```

style data histogram
set style histogram cluster gap 1
set style fill solid
set boxwidth
0.9 set xtics format ""
set grid ytics

set title "Week 1 January"
plot "barwlj.dat" using 2:xtic(1) title "Total Cycle Time" linecolor
    rgb green, \ '' using 3 title "Total Spindle Running Hours"
    linecolor rgb blue, \ '' using 4 title "Total Machine Ready
    Time" linecolor rgb purple, \ '' using 5 title "Total
    Breakdown Time" linecolor rgb red, \
    '' using 6 title "Total Machine Running(ON) Time" linecolor rgb peach

```

A.4 Program Code For Machine Utilization Bar Graph

```

    peach = "#FAEBD7"; green =
"#458b74"; set yrange [0:24] set style
data histogram
set style histogram
cluster gap 1 set style fill
solid
set boxwidth
0.9 set xtics format ""
set grid ytics

set title "Week 1 January"
plot "machutilizationlj.dat" using 2:xtic(1) title "Total Machine On Time" linecolor
    rgb peach, \ '' using 3 title "Successfully Used Time" linecolor
    rgb green

```

A.5 Comparison of Breakdown Time with respect to Total Machine Run Time

```

orange = "#fa8072"; set
    yrange [0:100]
set style data histogram
set style histogram cluster gap 1
set style fill solid
set boxwidth
0.9 set xtics format ""
set grid ytics

set title "Week 1 Inference"
plot "breakruncomparison.dat" using 2:xtic(1) title "Breakdown as a Percentage of Total Time" linecolor
    rgb orange, \

```

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PROJECT REPORT – 5

3. Publication

Cost Effective Digitalization Solution for Sinumerik CNC System To Increase The Transparency and Utilization of The Machine

Pooja Anand, Vinitha Lea Philip, Parthasarathy Eswaran

Abstract: *In this work, we fetch the current trends in industrial automation and data exchange technology adopted in Computer Numerical Control (CNC) machine and mitigate the features in a cost-effective manner. The current trend is Industry 4.0, uses cloud-based systems for information and data exchanges in machine to machine communication. This methodology is reliable, but expensive and can be afforded only by large scale companies. In order to provide the data transparencies at low cost, we utilize a low-cost computing system using Python language for small scale industry. This technique was implemented in the existing CNC machine and the machine parameters such as Machine Operating Mode, Cycle Time, Part Count, Feed rate, Spindle Running Hours, Machine Running Hours, and Machine Utilization Hours are monitored. Graphical user interface (GUI) screens are developed to help human machine interface. Acquired real-time machine data will help boost transparency and help the operator/ user for smart decision making. The IIoT (Industrial Internet of Things) technology helps to connect more numbers of such machines, results in increased machine utilization and productivity through continuously monitoring and analyzes.*

Index Terms: , IoT, Automation, Computer Numerical Control (CNC), Data analysis, Industry 4.0, Industrial IoT.

I. INTRODUCTION

Industry 4.0 is the fourth industrial revolution. It focuses on cyber physical systems, the Internet of Things, cloud computing and cognitive computing. This brings forth smart factories to the industrial world. Industry 4.0 has four design principles which include: Interconnection, Information transparency, Technical assistance and Decentralized decisions. A cost-effective system has been designed to measure the temperature and vibration variables of a machining process in Hass Computer Numerical Control (CNC)[1]. Industry 4.0 is revolution in a new wave of cyber-physical systems in NC (Numerically control) machining process platform which realizes the real-time monitoring and 3D display of machine tools[2-4].

Increasing efficiency has always been a major factor in the manufacturing sector for better production. Improved efficiency leads to better profitability. With the IoT gaining

importance, it has become one of the leading use cases for Industry 4.0[5], [6]. A combination of traditional condition monitoring enhanced with analytical algorithm forms the basis of Predictive maintenance strategies. Total Productive Maintenance (TPM) and 5S techniques minimize the breakdowns and improve the performance and efficiency of a machine [7]. This technology enables the prediction of machine failures before they occur. For many small business owners, the adoption of IoT may seem like a daunting challenge. Cloud computing based equipment monitoring systems help in monitoring the performance, statuses, equipment faults, production quality and precision of the machine [8]. These systems involve the use of expensive and complex software which are difficult to use. This bottleneck prevents many small scale industries from adopting these methods, and also their return on Investment takes a longer time. Monitoring machining processes have become a major factor for a manufacturer to improve the efficiency of the production line. Investigating the data of the CNC machine tool based on controller tuning operation help in increasing the productivity of industry 4.0[9-11]. It can also help in reducing the downtime of the machines. This work aims to help small manufacturing industries to use the current technology to improve functionality and identify the key areas for improvement and thereby increase the utilization by simplifying machine monitoring [12-13]. IoT technologies are the key factors in Industry 4.0 which help in increased product customization, productivity, and reliability of physical systems and are compared in real time [14]. Data extraction is made possible using industrial IoT in machines [15]. The proposed methodology will enhance small and medium enterprises to embrace IoT in a big way. These enterprises look for the following: cost should be affordable, the technology should be easy to use without any specialized knowledge or having to hire someone with special skills, it should be readily available, and the results should be accurate and must help them save or recover money faster. Our work is aimed to develop a solution that will help small and medium businesses achieve the above objectives. In our work, we will be monitoring the Sinumerik CNC with the help of a raspberry pi and thereby showing the machine's utilization patterns. The process of monitoring is initiated by selecting the list of parameters to be monitored [16]. The raspberry pi is programmed with the help of the programming language

Revised Manuscript Received on July 05, 2019.

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Python to meet the requirements.

Then the raspberry pi is hardwired to the Sinumerik CNC module to collect the required data over a specified period. The raspberry pi collects the data and collates for display. The various parameters that we require to be monitored are taken out from the CNC and given as input to the pi via the I/O card. The python program in the pi will run the proper algorithm to collect and store the data. The data is viewed on display screens in the form of graphical representations. The organization of this paper is as follows: Section II reviews the concepts of Industry 4.0 and the design of cost-effective interface Module, Section III details the experimental setup functionality, Graphical User Interface design and the implementation results was discussed in Section IV followed with conclusion.

II. BASIC CONCEPTS

The latest digital industrial technology is the Industry 4.0. This transformation makes it possible to analyze data of machines and thereby enabling faster and efficient production. Current market requirements and emerging autonomous technologies such as IoT are shifting the manufacturing companies' environment toward smart factories. Digitalization (Industry 4.0) is going to be a norm for all industries in the future. Most companies face challenges in adopting new technologies. In order to sustain a lead in the race, companies need to broaden in the field of digital technologies and implement digital manufacturing strategies. An Industry 4.0 solution will aid in overcoming the current challenges such as providing transparency, proper utilization of the machine, better production management, and thereby improving the efficiency of the manufacturing process.

A. Architecture

If we consider the first parameter, Auto/Manual mode, the process is as follows. There is a selector switch on the CNC screen where we select whether it is the auto mode or manual mode. The information is transferred to the memory of the CNC and stored as an NC variable which will be a digital value (0 or 1). The NC variable is stored in the CNC memory. In the PLC logic we will write a small logical code to access the data. The data available in the CNC memory will be transferred to the raspberry pi via the PLC. Every company has its proprietary PLC logic software, and for Siemens, it is called the Simatic Manager. In the Simatic manager, we will write a small logic to access the NC variable and bring it as data to store in the PLC. Siemens will do this programming. Another code is required to transfer the data from the PLC through the I/O card to the raspberry pi. The value will be taken from the system and moved like 0 or 1 through the I/O module. The SINUMERIK I/O Module is PP 72/48D 2/2A PN. It has 72 digital inputs and 48 digital outputs. The digital output is then connected to a relay board. Relays are used to avoid the risk of the raspberry pi burning out. The raspberry pi can only handle up to 5V, and the GPIOs can tolerate only

3.3V without relays. Then, data is be stored in the raspberry pi. Similarly, it will be done for all the parameters. The CNC was monitored over two weeks for around seven to eight hours, and this was done using the methods below.

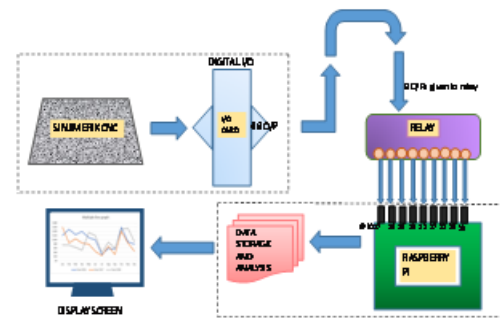


Fig. 1. Design of Cost-Effective digitization Module.

In the next section, we discuss the parameters selected. These parameters form the basis of our monitoring. More parameters can be added in the future with suitable additions to the programming.

B. Parameters

Machine Operating Mode: Auto/Manual: When the machine is in manual mode, the operator can push buttons, turn wheels, and turn switches on or off. In Auto mode, we execute the program that is fed into it.

Part Program Running: Yes/No: The set of instruction by which we can produce a part is known as a part program, and we can use to check the CNC program. This program gets executed when the cycle start button was pressed on the CNC.

Cycle Time: The time taken to finish a production run by the amount of fine products produced.

Part Count: The number of parts that have been produced during each production cycle is defined as the part count. It is monitored only when it runs in auto mode.

Feedrate Override: The feed-rate override is a multi-position switch which commonly ranges from 0 to 200 percent. It enables the setup person to slow (or stop) cutting motions on one end of the spectrum and double the programmed feed rate on the other.

Spindle Running Time: The spindle running hours is defined as the percentage of available time that the spindle of a machine is on.

Breakdown Hours: The breakdown hour is the amount of time when a system is unavailable or of time that a system fails to perform its primary functions.

Machine Running Hours: The machine running hours is the working of a machine for an hour. This is used as a basis for cost finding and for determining operating effectiveness.

Machine-Ready Time: Time taken after the part is produced until the next part is loaded onto the machine. It tells the operator that the machine is ready to start the process.

Machine Utilization Hours: It is the amount of time the machine is used successfully. Machine utilization compares the run time to the amount of time taken to setup the machine.

C. Software Tools

Tkinter tool: It is a toolkit for Python's GUI package. Tkinter is an inbuilt python



module. It is an object-oriented layer [21].

Python 3: Python is a high-level object-oriented scripting language that it is easily readable. The interpreter processes Python at runtime. The program does not need to be compiled before executing it[22].

Gnuplot: The Gnuplot is the tool used in our project to produce the graphs. Gnuplot programs help in generating two- and three-dimensional plots of functions [23].

III. EXPERIMENTAL SETUP

The I/O card connected to the CNC has 72 inputs and 48 outputs. Nine outputs are selected and given to the raspberry pi via the relay. Pins 17, 18,23,16,25,12,22,27, 24, 13 are provided as input for manual mode, auto mode, the part program running, cycle start, rejected part count, federate override greater than 100, spindle output, machine output, alarm status, and machine ready respectively. The raspberry pi is programmed using python to collect and analyze the data. Graphical user interface screens are designed in the form of tables and graphs for easy interaction.

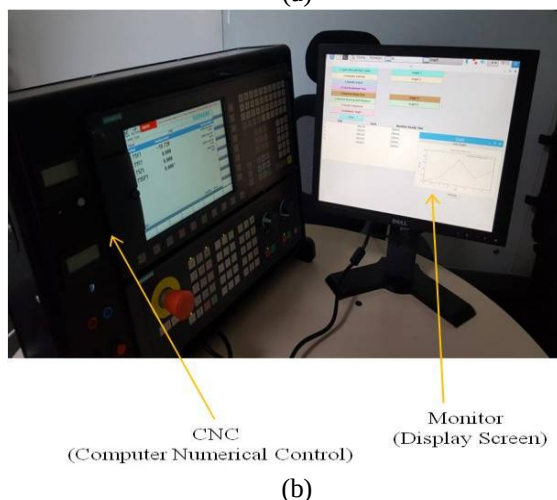
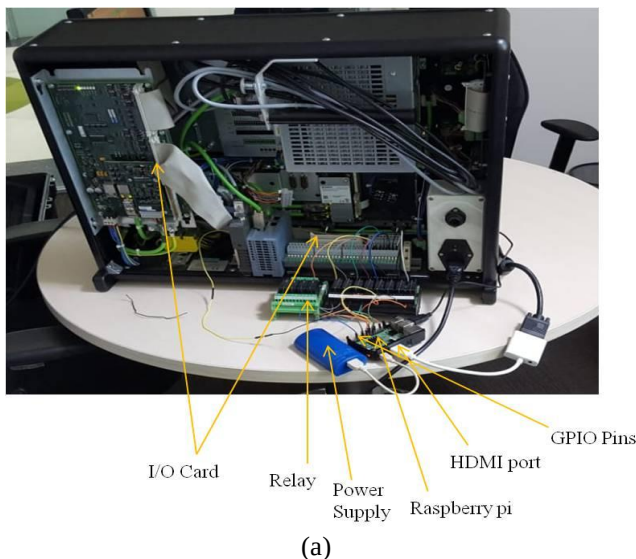


Fig 2. (a)Experimental setup back view, (b) Experimental Setup front view

A. Hardware Modules

Raspberry Pi 3: We have monitored the Sinumerik CNC with the help of raspberry pi. It is a mini computer that runs on the Linux platform and provides us with GPIO (General Purpose Input/Output) pins. It has 40 pins.

Sinumerik CNC: A CNC (Computer Numerical Control) is a device used for material removal to get desired parts/components. The Sinumerik CNC 828D is basically the NC Kernel with a built-in PLC in the front which is connected to an I/O card. This can be used to control many complex types of machinery including mills, lathes, and grinders.

I/O Module: The various parameters that have to be monitored are taken as an output from the CNC and given as input to the pi via the I/O card. The SINUMERIK I/O Module is PP 72/48D 2/2A PN. It has 72 digital inputs and 48 digital outputs. The digital output is connected to a relay board.

Relay: In our work, we need to connect the pi to a module with a higher voltage. Relays are used to avoid the risk of the raspberry pi burning out. The raspberry pi can only handle up to 5V, and the GPIOs can tolerate only 3.3V without relays. Relays have two main contacts NO and NC.

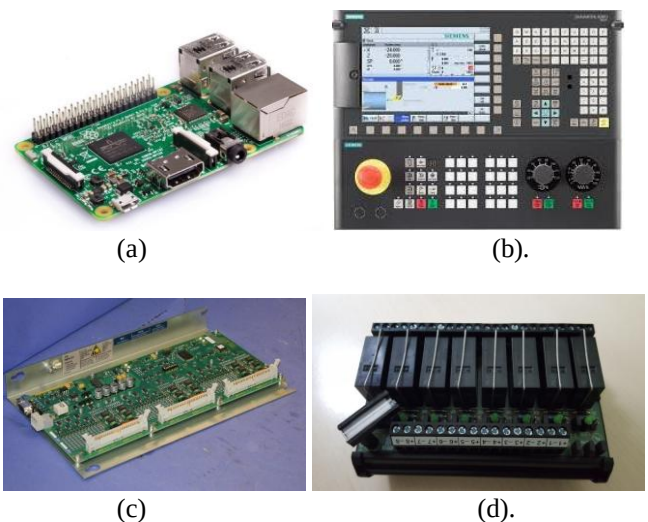


Fig. 3. (a). Raspberry Pi 3 Model B V1.2 [17], (b). Sinumerik 828 D [18], (c). Sinumerik I/O Module is PP 72/48D 2/2A PN [19], (d) 8 Channel Relay [20].

B. Software Design

The flow diagram for the respective parameters program logic is as follows:

Machine Operating Mode: AUTO/MANUAL and Part Program Running: YES/NO: In the raspberry pi, GPIO pin 18 and 17 are initialized as 'Auto Mode' and 'Manual Mode' respectively. The selected mode will be displayed. GPIO pin 23 is initialized as 'Part Program Selected'. The display will show whether the part program is running or not.

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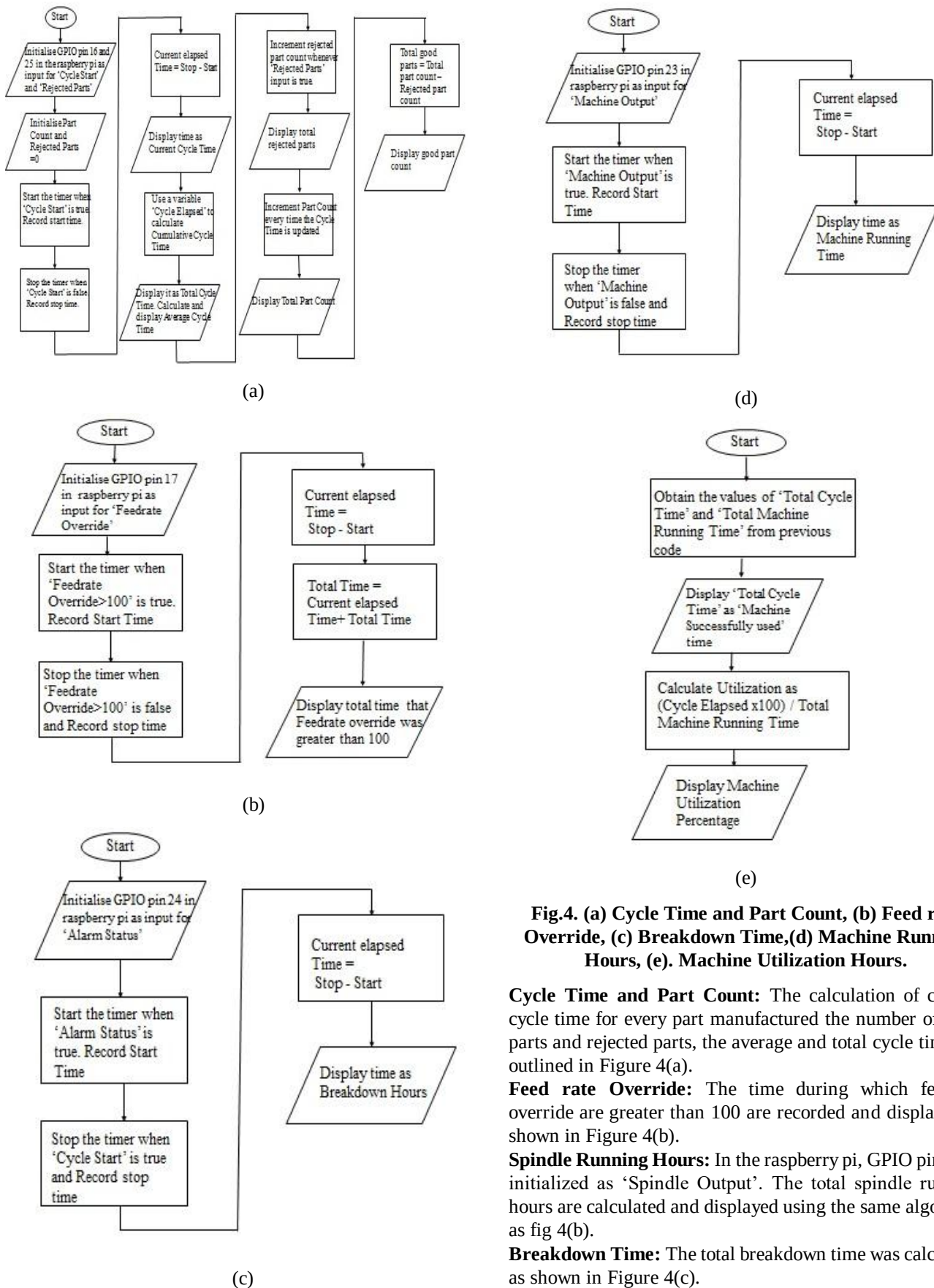


Fig.4. (a) Cycle Time and Part Count, (b) Feed rate Override, (c) Breakdown Time, (d) Machine Running Hours, (e). Machine Utilization Hours.

Cycle Time and Part Count: The calculation of current cycle time for every part manufactured the number of good parts and rejected parts, the average and total cycle time are outlined in Figure 4(a).

Feed rate Override: The time during which feedrate override are greater than 100 are recorded and displayed as shown in Figure 4(b).

Spindle Running Hours: In the raspberry pi, GPIO pin 22 is initialized as 'Spindle Output'. The total spindle running hours are calculated and displayed using the same algorithm as fig 4(b).

Breakdown Time: The total breakdown time was calculated as shown in Figure 4(c).

Machine-Ready: GPIO pin 23 was initialized as 'Machine Ready' and the total machine ready time is calculated using the same method as in fig 4(c).

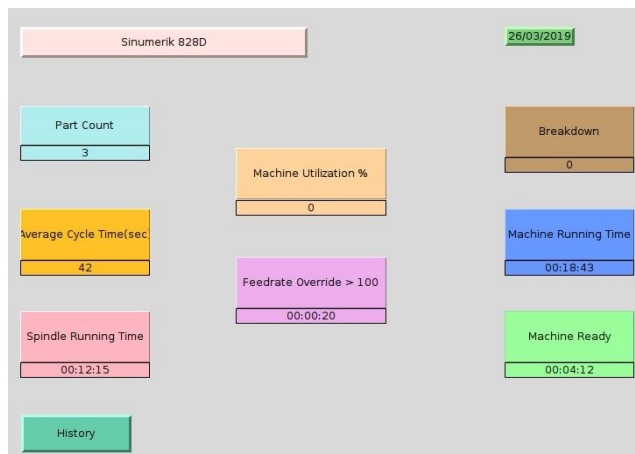
Machine Running Hours: Figure 4(d) demonstrates how

the total machine running time was calculated.

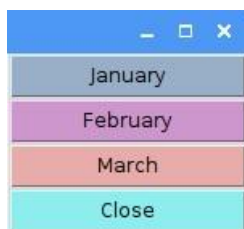
Machine utilization hours: The machine utilization percentage is calculated as shown in Figure 4(e).

C. Graphical User Interface Module

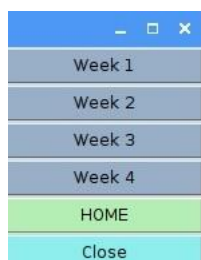
We have designed a Machine Utilization Dash Board which is displayed below. This is a system to help the operator to understand the production status of the machine at the current time. The information is given in the form of graphs, signals, numbers and lights to alert the user about the issues. This makes the state and condition of the processes easily accessible and clear to everyone. This helps the industry in monitoring and hence improving productivity.



(a)



(b)



(c)

Fig 5. (a) Machine Utilization Dashboard, (b).GUI of Months, (c) GUI of Weeks.

When the history button in the dashboard is clicked, it shows screens that display the past data as required. The first screen displays the months. On selecting January, February, and March, the respective weeks are displayed.

IV. RESULTS AND DISCUSSION

On selecting a week, the parameters will pop up. The same list of parameters will be displayed irrespective of which week is selected. On selecting each parameter, the corresponding table will be displayed along with the on monitoring the parameters over two weeks (eight hours daily), we have the following data as shown in the tables. The data recorded in the first week of January is shown below:

Cycle time and Part Count: It is inferred from Figure 7(a) that the average time for 1 part to be produced in a day is 48 sec. The total number of parts produced in a day is 425 parts and the total time for 425 parts to be produced is 5hrs 40mins 0sec.

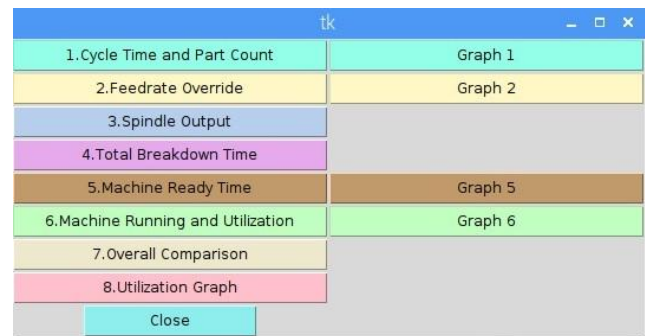


Fig 6. GUI Screen of Parameters

Cycle Time and Part Count				
Day	Date	Average Cycle Time	Total Cycle Time	Part Count
1	1/01/19	48sec	5hr 40min 0sec	425
2	2/01/19	50sec	5hr 45min 0sec	414
3	3/01/19	48sec	5hr 40min 48sec	426
4	4/01/19	51sec	5hr 48min 30sec	410
5	7/01/19	49sec	5hr 43min 49sec	421

(a)

Feedrate Override			Total Spindle Running Hours		
Day	Date	Feedrate Override >100	Day	Date	Total Spindle Running Hours
1	1/01/19	25min	1	1/01/19	6hr 20min 0sec
2	2/01/19	55min	2	2/01/19	6hr 25min 0sec
3	3/01/19	50min	3	3/01/19	6hr 20min 48sec
4	4/01/19	45min	4	4/01/19	6hr 28min 30sec
5	7/01/19	20min	5	7/01/19	6hr 23min 49sec

(b)

(c)

Breakdown Hours			Total Machine Ready Time		
Day	Date	Breakdown Hours	Day	Date	Total Machine Ready Time
1	1/01/19	1hr 15min 0sec	1	1/01/19	10mins
2	2/01/19	0hr 35min 0sec	2	2/01/19	12mins
3	3/01/19	1hr 20min 0sec	3	3/01/19	18mins
4	4/01/19	0hr 48min 0sec	4	4/01/19	13mins
5	7/01/19	0hr 40min 0sec	5	7/01/19	16mins

(d)

(e)

Machine Running time and Utilization Percentage			
Day	Date	Total Machine Running Time	Machine Utilization Percentage
1	1/01/19	6hr 45min 10sec	86.3 %
2	2/01/19	6hr 46min 18sec	85.4 %
3	3/01/19	6hr 51min 23sec	87.3 %
4	4/01/19	6hr 47min 28sec	82.6 %
5	7/01/19	6hr 43min 16sec	85.1 %

(f)

Fig 7. (a) Cycle time and Part Count, (b) Feed rate Override, (c) Spindle Running Time, (d) Breakdown Time, (e) Machine Ready Time, (f). Machine Running and Utilization Percentage.

Feed rate Override: Figure 7(b) depicts that the feed rate override was greater than 100 for 25mins on 1st January 19. This shows that the operator has increased the programmed feed rate of over 100 for 25mins.

Spindle Running Time: Figure 7(c) shows the amount of time the spindle has functioned for each day in a week. The total spindle running hours for one day is 5hrs 80mins 0sec.

Breakdown Time It can be inferred from Figure 7(d) the amount of time in a day when the machine was unavailable or failed to perform its functions. The total



Cost Effective Digitalization Solution for Sinumerik CNC System To Increase The Transparency and Utilization of The Machine

breakdown time in a day is 1hr 15min 0sec.

Machine-Ready Time: The above Figure 7(e) shows the time taken after a part is produced until the next part is loaded onto the machine. The total machine ready time for one day is 10mins.

Machine Running and Utilization Percentage: The total machine running time in a day was found to be 6hr 45min 10sec from Figure 7(f). The total machine utilization percentage for one day is 86.3%.

The first graph is a comparison of total cycle time, total spindle running hours, total machine ready time, breakdown time and total machine running (ON) time. The second graph is a comparison between successfully used time and total machine running time which gives us an understanding of the machine utilization. The graphical representation of the parameters data is depicted and shown in Figure 8:

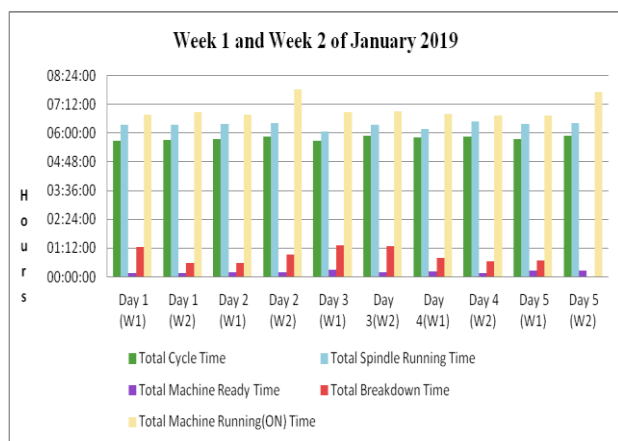


Fig 8. Comparison of various parameters

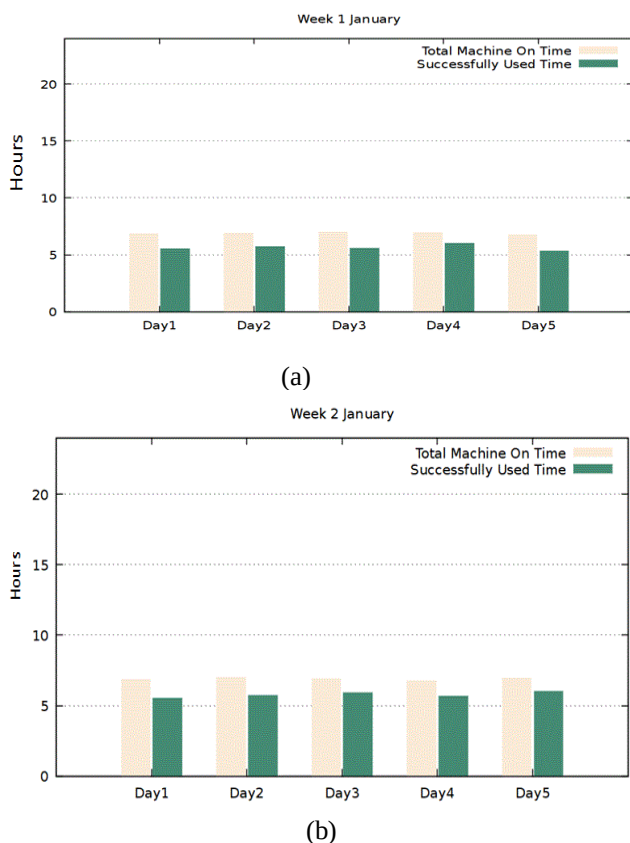


Fig 9. (a) Machine Utilization Graph for Week 1, (b). Machine Utilization Graph for Week 2.

Similarly, the graphical representation for the data recorded in the second week of January is depicted in Figure 9.

On analyzing the data, it is evident that breakdown time is significantly affecting the production. This graph shows how much percentage of the total time was wasted in the breakdown.

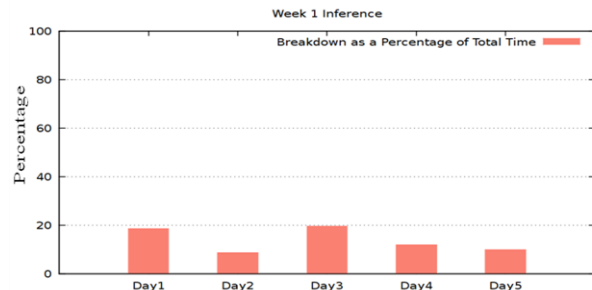


Fig 10. Breakdown time in Percentage of Total Time.

Proper maintenance is essential for extending the life of the machine and increasing productivity. Despite this, for small manufacturing enterprises, there are not much tools or equipment available to understand the impact of the machine breakdown or production loss, and rarely are these variables measured. We intend to provide the correlation between the actual machine running time and the successfully utilized time so that we will know to what extent there are losses and this will help in better planning of maintenance and future activities. The tables and graphs help us in visualizing the impact of various parameters. We aim to achieve maximum efficiency by improving utilization. One way to reduce breakdowns is to ensure proper alignment of all moving components and proper lubrication and cooling systems. When a product is getting machined, if at that point of time the machine breaks down, there are chances that we may not be able to reuse that particular piece. In some cases, we may have to start with a fresh piece, which results in a loss. In this particular instance, we were able to continue producing the part even after the breakdown, but in other cases, it may be required to restart the process. Minimization of breakdown time can be achieved by scheduling proper maintenance. We should also make sure that the ready time should be nominal and minimize defective part count. The design of CCEMS for the CNC machine tool and NWAIF gives many benefits [8] but, here in our work, we have focused on a solution independent of the cloud. The cycle time per piece has been monitored along with the spindle speed for Hass CNC and data is stored in the cloud with the focus mainly on maintenance [1]. In our work, we have monitored various parameters, analyzed and stored the data with the same accuracy and then display them in the form of tables and graphs without using the internet; hence our method is cost effective.

V. CONCLUSION

Smart factories take the manufacturing industries a leap forward from traditional automation to a fully connected and flexible system, which compels the companies to take up the latest industrial mechanisms. We have provided a feasible, cost-effective solution using a raspberry pi to simulate an Industry 4.0 solution for CNC. This is a solution for small manufacturing companies to adopt new technologies for improving overall efficiency and become more competitive. We can capture the machine utilization parameters easily over a weekly period and simulate the acquired data in graphical form with the help of user interface screens. Data acquisition is done in real time so that the user can analyze the performance of the machine and the production rate at the current time. This method increases transparency, thereby giving insight on where the scope is available to improve machine utilization. This helps the user to get more profit, production, and higher efficiency. Adopting cost-effective technology for monitoring and managing the utilization and efficiency of machine tools will help in reducing waste and becoming more productive.

VI. ACKNOWLEDGMENT

We would like to express our gratitude towards Mr.K.K.Vivek (Service Operations Team Leader (CHN and CBE)) Siemens Ltd , Mr.Joseph S (Vertical (Indirect) Sales Professional (CHN)) Siemens Ltd for giving us an opportunity to work at Siemens Ltd and for their guidance and support which helped us in completion of this work.

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Cost Effective Digitalization Solution for Sinumerik CNC System To Increase The Transparency and Utilization of The Machine



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PROJECT REPORT – 5

4. Evaluation Rubrics

SRM Institute of Science & Technology
College of Engineering and Technology
Department of Electronics and Communication Engineering

EVALUATION PROCESS TO IDENTIFY BEST AND AVERAGE PROJECTS

The Major project is assessed and evaluated based on Program Outcomes achievement which covers Problem analysis, Design component, Investigation Methodology, Usage of contemporary tools, Project management and Presentation. Best and average project are assessed using evaluation rubrics applied on Project Report, Presentation and Demonstration.

A. The Project Work will be assessed using the Assessment Rubrics given below

- Project goals and problems are clearly identified. The chosen solution was well thought of.
- Design strategy development which includes, plan to solve the problem, decomposition of work into subtasks, and development of a timeline using Gantt chart.
- The implementation (also problem solving) is very systematic. Proper assumptions made; results are correctly analysed and interpreted.
- Properly choose and correctly use all the techniques, skills, and modern engineering tools for their project.
- Understanding on the impact of engineering solutions in a global, economic, environmental, and societal context and he/she provides an in-depth discussion.
- Deep understanding of the professional issues involved and the ethical implications of the project, system, etc.
- Information is presented in a logical, interesting way, which is easy to follow. Purpose is clearly stated and explains the structure of work.
- Student can demonstrate effective project management skills and problem solving techniques related to project management. Can apply the management principles such as cost benefit analysis, strategic alignment and project portfolio management and project performance analysis and metrics. Can deliver successful projects at a faster pace in increasingly complex environments. Can demonstrate a strong understanding of project finance and the various metrics associated with the monitoring of the financial health of the project.
- Capability of doing research on his/her own, i.e. he/she can do a complete research related to the project.

B. Project Report is assessed based on the assessment rubrics given in Table 1.

Table 1: Project Report Assessment Rubrics

Particulars	Exceptional
Objective	Objective complete and well-written; provides all necessary background principles for the experiment
Content	✓ Technically correct ✓ Contain in-depth and complete details of the project ✓ An engineer can recreate the project based on the report.
Language (Word Choice, Grammar)	✓ Sentences are complete and grammatical. They flow together easily ✓ Words are chosen for their precise meaning. ✓ Engineering terms and jargon are used correctly. ✓ No misspelled words.
Experimental procedure	Well-written in paragraph format, all experimental details are covered
Numerical Usage and Illustrations	✓ All figures, graphs, charts, and drawings are accurate, consistent with the text, and of good quality. They enhance understanding of the text. ✓ All items are labeled and referred to in the text. ✓ All equations are clear, accurate, and labeled. All variables are defined and units specified. Discussion about the equation development and use is stated.
Results, Discussion and Conclusions	✓ All important trends and data comparisons have been interpreted correctly and discussed, good understanding of results is conveyed. ✓ All important conclusions have been clearly made, student shows good understanding
Visual Format and Organization	✓ Structuring the content to represent the logical progression ✓ The doc. is visually appealing and easily navigated. ✓ Usage of white space is used as appropriate to separate blocks of text and add emphasis.
Use of references	✓ Prior work is acknowledged by referring to sources for theories, assumptions, quotations, and findings. ✓ Correct information for References.
Realistic constraints	✓ Incorporates appropriate multiple realistic constraints such as economic, environmental, social, political, ethical, health and safety, manufacturability, and sustainability ✓ Analysis provides correct reasons as how this constraint affects the design of the system, component, or process and contains in-depth discussion.
Engineering Standards	Clear evidence of ability to use engineering principles to design components, devices or systems

C. Project Presentation is assessed based on the assessment rubrics given in Table 2.

Table 2: Project Presentation Assessment Rubrics

Particulars	Exceptional
Content	✓ Presentation contains all required components ✓ A complete explanation of major concepts and theories is provided and drawn upon relevant literature ✓ Content is consistently accurate
Organization	✓ Presentation is clear, logical and organized ✓ Audience can follow line of reasoning
Professional delivery	✓ Presenters are comfortable in front of audience and his/her voice is audible ✓ No reading from the notes or presentation ✓ Sentences are complete and grammatical, and they flow together easily
Visual Aids	✓ ability to understand the message ✓ grammar and choice of words
Conclusion of presentation	✓ Planned concluding remarks (not just “I guess that’s it.”) ✓ Presented significant results
Responses to questions	✓ Listened to questions without interrupting ✓ Began with general answer and then followed up with details

D. Project Demonstration is assessed based on the assessment rubrics given in Table 3.

Table 3: Project Demonstration Assessment Rubrics

Particulars	Exceptional
Introduction	✓ Clearly identifies and discusses focus/purpose of project. ✓ A complete explanation of major concepts and theories is provided and drawn upon relevant literature.
Methodology	✓ Presented the detailed design, including modelling, control design, simulation, and experimental results, with diagrams and parameter values. ✓ Compared simulation and experimental results. Compared achieved performance with the design specification. ✓ Provided solid technical data, and presented it in an easily grasped manner, using graphs where possible.
Organization & Presentation	✓ Have all the materials required for the project demonstration ✓ All these materials are neatly organized so that the demonstration runs smoothly ✓ Speech, confidence, knowledge and enthusiasm are inspirational ✓ Good eye contact and voice projection maintained throughout the

	entire presentation ✓ Group understands what they are doing and carries out the demonstration as planned in an enthusiastic manner. There is a very good understanding of the "how and why" of the project
Interest/Excitement	✓ Demonstration was very interesting and captured the excitement of all those viewing the presentation.
Professionalism	✓ Respectable at all times. Shows extensive practice and preparation. No safety issues during demonstration.
Social Impact and Authenticity	✓ The project has an authentic context, involves real-world tasks, tools, and quality standards, and makes a real impact on the world.
Realistic constraints	✓ Incorporates appropriate multiple realistic constraints such as economic, environmental, social, political, ethical, health and safety, manufacturability, and sustainability. ✓ Analysis provides correct reasons as how this constraint affects the design of the system, component, or process and contains in-depth discussion.
Engineering Standards	✓ Incorporates appropriate engineering standards that defines the characteristics of a product, process or service, such as dimensions, safety aspects, and performance requirements.
Results, Discussion & Conclusion	✓ Results are clearly explained in a comprehensive level of detail and are well-organized. ✓ Interpretations/ analysis of results are thoughtful and insightful ✓ Suggestions for further research in this area are provided and are appropriate

E. Publications

Students are encouraged to publish their contribution of major project outcomes in reputed indexed or non-indexed journals/ conferences. Based on their publication the outcome of the project work is gauged. **Students are advised to publish their research articles in Scopus/SCI indexed Journals.**

F. Best Practices in Major Project:

COMSPRO is the Major Project Design contest conducted every year in the department to showcase the top 3 projects chosen from each domain by the respective project coordinators, to the pre-final and second year students to motivate them to improve their design skills. Judges were identified for the COMSPRO and were asked to select the winners of the contest. The purpose of this design contest is to **increase the student motivation, engagement, confidence, self-perceptions and demonstration of the learning proficiency.** The preparatory work involved in the conduction of COMSPRO for the remaining years say AY 2018-19 and 2017-18 are as follows:

- **COMSPRO banner for wide publicity**
- **Evaluation Criteria for Judges**
- **Announcement of Winners**
- **Certificate for Best Project Award**

PROJECT REPORT – 5

5. Assessment record for Review 1, 2, 3 and CO & PO Mapping

Major Project - First review Nov-23 - Fully Internship																
Batch No	Register.No	Student Names	Project Title	Proj. Guide	Novelty(5)			Methodology(5)			PPT(5)			Guide(5)	Total(10)	
					P1	P2	P3	P1	P2	P3	P1	P2	P3			
1	RA1511004010534	Devyani Sharma(lead)squared)	railway track crack detection	Mrs. S. Hannah Pauline	4	4	4	4	4	4	4	4	4	4	8	
	RA1511004010176	Akshat singh(lead)squared)			4	4	4	4	4	4	4	4	4	8		
	RA1511004010482	aditi kothari			5	5	5	4	4	4	4	4	4	9.3		
	RA1511004010693	Deepesh Acharya(lead)squared)	4	4	4	4	4	4	4	4	4	4	8			
	RA1511004010608	Anjali	4	4	4	4	4	4	4	4	4	5	9			
	RA1511004010622	Ayush Dhawan	"Bidirectional convertor for electric bike with Charging Feature	Mr.A Josua Jefferson.	4	4	4	5	5	4	4	4	4	5	9.2	
2	RA1511004010626	Shreya Singh	Automated Car Parking System	Dr.P. Aruna Priya	5	5	4	5	4	5	4	5	4	5	9.5	
RA1511004010728	Yagnya	4			5	5	5	5	5	4	5	4	8.75			
3	RA1511004010205	Prajwal Amar Singh			5	5	4	4	5	4	4	4	4	4	8.3	
4	RA1511004010119	P. Sai Sisira	V380 wireless camara-home security and remote intelligence baby care	Mr. B. Viswanathan	4	4	4	4	5	4	4	4	4	5	9.1	
5	RA1511004010196	B Arun Kumar	Data path processing design in physical layer for LTE	AVM MANIKANDAN	4	4	4	4	4	4	4	4	4	4	8	
6	RA1511004010531	Karthik Subramanian	Development of channel decoding modules	Dr. P. Aruna Priya	5	4	5	4	4	4	4	4	5	5	9.3	
7	RA1511004010268	M Jagath	Camera systems	Maria Dominic Savio	4	4	4	3	4	4	3	3	3	3	6.5	
8	RA1511004010095	Y Shanmukh Chowdary			4	4	4	3	4	4	3	3	3	3	6.5	
9	RA1511004010131	Sai Krishna S	Design of hardware accelerators processed in	Dr. P. Aruna Priya	5	5	5	5	5	5	5	4	5	5	4	8.8
10	RA1511004010128	Anurup Ojha	Design and Implementation of Telemetry encoder for space applications	Dr.B Ramachandran	4	5	4	4	5	4	4	5	4	4	4	8.3
11	RA1511004010543	Abhishek Mazumdar	Design and Simulation Studies of Multioctave Antipodal Vivaldi Antenna in Electronic Warfare Applications	Mr A.V.M Manikandan	4	4	5	4	4	4	5	4	4	4	4	8.2
12	RA1511004010595	K.Rohit	VLSI design and Verification of multichannel ADC/DAC Controller Unit for RedPine SoC	Dr.J.Selvakumar	4	4	4	4	4	4	4	4	4	4	4	8
13	RA1511004010380	Anirudh Sunil Warrior	High speed sampling and storage of analog wave	Dr.B Ramachandran	4	4	5	5	4	4	4	4	5	5	5	9.3
14	RA1511004010601	Vasushree Goyal	Design and development of industrial grade	Dr. P. Eswaran	4	4	5	5	4	4	4	4	4	3	7.2	
15	RA1511004010695	Ridhima Bahl	Phase and time delay estimation and correction	Dr. C.T. Manimegalai	4	5	4	5	5	5	5	4	5	5	5	9.6
Batch No	Register.No	Student Names	Project Title	Proj. Guide	Novelty(5)			Methodology(5)			PPT(5)			Guide(5)		
					P1	P2	P3	P1	P2	P3	P1	P2	P3			
15	RA1511004010030	M. Pranav	Open Switch Based API on Virtual Machines	Mr. U. Hari	4	4	4	5	4	5	4	4	4	4	5	9.2
	RA1511004010042	S. Srikanth			5	4	5	5	5	5	4	4	4	5	9.5	
	RA1511004010122	C.V.K. Anirudh Jagannath			4	4	4	4	4	4	5	5	5	5	9.3	
	RA1511004010107	Pooja Anand			5	4	5	5	4	5	5	4	5	5	9.65	
16	RA1511004010059	Vinitha Lea Philip	Solution for Siemens Sinumerik CNC System	Dr. P. Eswaran	5	4	5	4	5	4	4	4	4	5	9.3	
17	RA1511004010384	Ashrai Jha	Modified gate level computing method for complexity reduction in CSE technique	Dr. Selvakumar	4	4	4	3	3	4	4	3	4	3	6.6	
	RA1511004010333	Deepansh madan			4	4	4	4	3	3	3	3	3	4	4	7.5
	RA1511004010377	Anshul Tayal			4	4	4	3	3	3	3	3	3	4	4	7.4
	RA1511004010377	Anshul Tayal			4	4	4	3	3	3	3	3	3	4	4	7.4

SRM INSTITUTE OF SCIENCE AND TECHNOLOGY

Department of ECE

Major Project - Semester 7 review marks - Fully Internship

Batch No	Register.No	Student Names	Project Title	Proj. Guide	Content Delivery (10)			Partial result/output(5)			Implementation(5)			PPT(5)			Guide(5)	Guide(25)	Average(25)	Total (30)
					P1	P2	P3	P1	P2	P3	P1	P2	P3	P1	P2	P3				
1	RA1511004010534	Devyani Sharma(leadssquared)	railway track crack detection	Mrs. S. Hannah Pauline	9	9	9	5	4	5	4	5	5	4	4	4	5	25	23	14
	RA1511004010176	Akshat singh(leadssquared)			8	8	8	4	4	4	4	5	4	4	4	4	5	24	21.25	13.25
	RA1511004010482	aditi kothari			8	8	8	5	4	4	4	4	4	4	4	4	5	24	21.25	13.25
	RA1511004010693	Deepesh Acharya(leadssquared)			8	8	8	4	4	4	4	4	5	4	4	4	5	24	21.25	13.25
	RA1511004010608	Anjali	"Bidirectional convertor for electric bike with Charging Feature"	Mr.A Josua Jefferson.	10	9	9	4	4	4	5	5	5	4	3	3	4	24	22.25	13.25
2	RA1511004010622	Ayush Dhawan			9	9	9	4	4	4	4	5	5	4	3	3	4	24	21.75	13
	RA1511004010626	Shreya Singh			10	9	9	4	4	4	5	5	5	4	3	3	4	24	22.25	13.25
3	RA1511004010205	Prajwal Amar Singh	Automated Car Parking System	Dr.P. Aruna Priya	8	7	7	3	3	3	3	2	3	4	3	3	4	18	16.75	10.5
4	RA1511004010119	P. Sai Sisira	Home security and baby monitoring using arduino	Mr. B. Viswanathan	8	7	7	4	4	4	4	4	4	4	4	3	4	20	19.25	11.75
5	RA1511004010196	B Arun Kumar	Data path processing design in physical layer for LTE	AVM MANIKANDAN	8	7	7	5	4	4	5	4	4	5	4	4	4	24	22.25	13.25
6	RA1511004010531	Karthik Subramanian	Development of channel decoding modules	Dr. P. Aruna Priya	9	9	9	4	3	3	4	4	4	5	4	4	4	24	22.5	13.25
7	RA1511004010268	M Jagath	Conversion from 2D chip sensor to 3D chip sensor in endoscopic camera systems	Maria Dominic Savio	8	8	9	4	4	4	4	3	4	5	4	5	4	22	21	12.5
8	RA1511004010095	Y Shanmukh Chowdary	Design of hardware accelerators processed in	Dr. P. Aruna Priya	9	9	9	5	5	4	5	4	4	4	4	4	4	22	22	13
9	RA1511004010131	Sai Krishna S	Design and Implementation of Telemetry encoder for space applications	Dr.B Ramachandran	10	10	9	5	5	4	5	5	5	5	5	4	4	20	23	13.5
10	RA1511004010128	Anurup Ojha	Design and Simulation Studies of Multioctave Antipodal Vivaldi Antenna in Electronic Warfare Applications	Mr A.V.M Manikandan	9	8	8	5	5	5	4	4	4	5	4	4	4	23	23	13.5
11	RA1511004010543	Abhishek Mazumdar	VLSD design and Verification of multichannel ADC/DAC Controller Unit for RedPine SoC	Dr.J.Selvakumar	8	7	8	4	3	4	4	3	4	4	4	3	3	18	18.5	10.75
12	RA1511004010595	K.Rohit	High speed sampling and storage of analog	Dr.B Ramachandran	8	8	8	4	4	4	3	3	4	4	3	3	4	18	18.5	11.25
13	RA1511004010380	Anirudh Sunil Warrior	Design and development of industrial grade	Dr. P. Eswaran	9	7	8	4	4	4	4	4	4	4	3	3	4	16	18.5	11.25
14	RA1511004010601	Vasushree Goyal	Phase and time delay estimation and correction	Dr. C.T. Manimegalai	10	9	8	4	5	4	5	4	4	5	4	4	5	24	22.5	13.75
15	RA1511004010059	Ridhima Bahl																		
16	RA1511004010030	M. Pranav	Open Switch Based API on Virtual Machines	Mr. U. Hari	10	10	10	5	5	4	5	5	4	5	5	4	5	25	24.25	14.75
	RA1511004010042	S. Srikanth			10	10	9	5	5	5	5	5	4	5	5	4	5	25	24.25	14.75
17	RA1511004010122	C.V.K. Anirudh Jagannath	Low Cost Digitalization (Industry 4.0)	Dr. P. Eswaran	10	10	9	5	5	5	5	5	5	5	5	5	5	25	24.75	15
	RA1511004010107	Pooja Anand			10	10	10	5	5	5	4	5	5	5	5	5	5	23	24.25	14.75
18	RA1511004010059	Vinitha Lea Philip	Solution for Siemens Sinumerik CNC System	Dr. P. Eswaran	10	10	10	5	5	5	4	5	5	5	5	5	5	23	24.25	14.75
	RA1511004010384	Ashrai Jha			8	8	8	5	5	4	4	4	4	5	4	4	4	20	20.75	12.5
19	RA1511004010333	Deepansh madan	Modified gate level computing method for complexity reduction in CSE technique	Dr. Selvakumar	9	9	8	4	4	4	5	5	4	5	5	5	4	20	21.75	13
	RA1511004010277	Anshul Tayal			9	9	8	5	5	4	5	5	4	5	5	4	4	23	22.75	13.5

Academic Advisor

Project coordinator

SRM INSTITUTE OF SCIENCE AND TECHNOLOGY
Department of ECE
Major Project - Third rev. Marks - Fully Internship

Batch No	Register.No	Student Names	Project Title	Proj. Guide	Presentation (50)				Presentation	Poster (150)				Poster (30)	Project report	Project report (25)	Total (80)	Final (20)
					P1	P2	PC	G		P1	P2	PC	G					
	RA1511004010176	Akshat singh(lead squared)	Grocery store automation using deep learning	Mrs. S. Hannah Pauline	46	45	45	50	23.3	137	138	138	150	28.15	45	22.5	73.9	18.475
	RA1511004010482	aditi kothari			45	46	46	50	23.4	137	138	138	150	28.15	45	22.5	74.025	18.5063
	RA1511004010534	Devayani			45	44	44	50	22.9	137	138	138	150	28.15	45	22.5	73.525	18.3813
	RA1511004010693	Deepesh Acharya(lead squared)			46	45	45	50	23.3	137	138	138	150	28.15	45	22.5	73.9	18.475
	RA1511004010608	Anjali	"Bidirectional convertor for electric bike with Charging Feature"	Mr.A Josua Jefferson.	45	44	44	47	22.5	140	142	142	144	28.4	45	22.5	73.4	18.35
	RA1511004010622	Ayush Dhawan			45	46	46	47	23	140	142	142	144	28.4	45	22.5	73.9	18.475
2	RA1511004010626	Shreya Singh			45	45	45	47	22.8	140	142	142	144	28.4	45	22.5	73.65	18.4125
3	RA1511004010205	Prajwal Amar Singh	Automated Car Parking System	Dr.P. Aruna Priya	44	39	40	41	20.5	129	125	125	130	25.45	40	20	65.95	16.4875
4	RA1511004010119	P. Sai Sisira	Home security and baby monitoring using arduino	Mr. B. Viswanathan	44	44	44	45	22.1	129	135	135	128	26.35	45	22.5	70.975	17.7438
5	RA1511004010196	B Arun Kumar	Data path processing design in physical layer for LTE	AVM MANIKANDAN	46	46	46	48	23.3	137	140	140	142	27.95	47	23.5	74.7	18.675
6	RA1511004010531	Karthik Subramanian	Development of channel decoding modules	Dr. P. Aruna Priya	48	48	46	48	23.8	138	141	140	142	28.05	50	25	76.8	19.2
7	RA1511004010268	M Jagath	Conversion from 2D chip sensor to 3D chip sensor in endoscopic camera systems	Maria Dominic Savio	44	42	44	42	21.5	139	135	136	130	27	42	21	69.5	17.375
	RA1511004010095	Y Shanmukh Chowdary			43	42	44	42	21.4	139	135	136	130	27	42	21	69.375	17.3438
8	RA1511004010131	Sai Krishna S	Design of hardware accelerators processed	Dr. P. Aruna Priya	44	48	46	46	23	143	146	142	144	28.75	48	24	75.75	18.9375
9	RA1511004010128	Anurup Ojha	Design and Implementation of Telemetry encoder for space applications	Dr.B Ramachandran	45	45	45	42	22.1	127	130	132	134	26.15	46	23	71.275	17.8188
10	RA1511004010543	Abhishek Mazumdar	Design and Simulation Studies of Multioctave Antipodal Vivaldi Antenna in Electronic Warfare Applications	Mr A.V.M Manikandan	44	46	46	50	23.3	138	135	135	138	27.3	44	22	72.55	18.1375
11	RA1511004010595	K.Rohit	VLSI design and Verification of multichannel ADC/DAC Controller Unit for RedPine SoC	Dr.J.Selvakumar	46	47	46	48	23.4	140	136	135	142	27.65	48	24	75.025	18.7563
12	RA1511004010380	Anirudh Sunil Warriar	High speed sampling and storage of analog v	Dr.B Ramachandran	40	41	40	40	20.1	105	110	110	105	21.5	32	16	57.625	14.4063
13	RA1511004010601	Vasushree Goyal	Design and development of industrial grade	Dr. P. Eswaran	44	44	44	44	22	138	138	138	118	26.6	33	16.5	65.1	16.275
14	RA1511004010695	Ridhima Bahl	Phase and time delay estimation and correction	Dr. C.T. Manimegalai	48	46	46	48	23.5	141	142	142	146	28.55	50	25	77.05	19.2625
Batch	Register.No	Student Names	Project Title	Proj. Guide														
15	RA1511004010030	M. Pranav	Open Switch Based API on Virtual Machines	Mr. U. Hari	46	46	47	49	23.5	138	138	138	143	27.85	50	25	76.35	19.0875
	RA1511004010042	S. Srikanth			45	46	45	48	23	138	138	138	143	27.85	50	25	75.85	18.9625
	RA1511004010122	C.V.K. Anirudh Jagannath			46	46	45	50	23.4	138	138	138	143	27.85	50	25	76.225	19.0563
	RA1511004010107	Pooja Anand			48	48	50	50	24.5	140	144	144	146	28.7	44	22	75.2	18.8
16	RA1511004010059	Vinitha Lea Philip	Solution for Siemens Sinumerik CNC System	Dr. P. Eswaran	48	48	48	49	24.1	140	144	144	146	28.7	44	22	74.825	18.7063
17	RA1511004010384	Ashrai Jha	Modified gate level computing method for complexity reduction in CSE technique	Dr. Selvakumar	45	44	44	43	22	139	144	141	131	27.75	46	23	72.75	18.1875
	RA1511004010333	Deepansh madan			45	46	46	45	22.8	139	144	141	131	27.75	46	23	73.5	18.375
	RA1511004010277	Anshul Tayal			45	44	44	44	22.1	139	144	141	131	27.75	46	23	72.875	18.2188

(Signature)
Project coordinator

(Signature)
Professor incharge

15EC496L - PROJECTWORK

PresentationAssessmentRubric(tobefilledbythereviewers)

TitleoftheProject: Low Cost Digitalization (Industry 4.0) Solution for Siemens Sinumerik CNC System to Increase the Transparency and Utilization of the Machine

Presenter(s)Name: Pooja Anand (RA1511004010107) & Vinitha Lea Philip(RA1511004010059)
(Write Reg.No. & Class within braces for each student member)

ReviewNo. 3

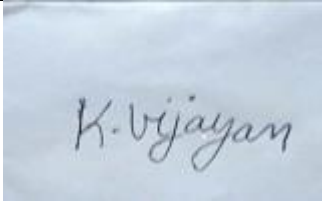
Date: 8/5/2019

Supervisor(s)Name&Designation: Dr.P.Eswaran Associate professor

Rubric	Unacceptable (1)	Marginal (2)	Acceptable (3-4)	Exceptional (5)	Score of the Presenters											
					RA1511004010107				RA1511004010059							
					P1	P2	PC	G	P1	P2	PC	G	P1	P2	PC	G
Content: Components (Introduction, Problem statement, Design, Schedule, Cost,Summary)	Presentation is missing several major required components. Has no understanding of the design problem.	Presentation is missing a few required components. Shows understanding of the design problem. Lacks alternative design concepts.	Presentation contains most of required components. Understands the design problem and objective. Has alternative designs concepts been investigated.	Presentation contains all required components. Shows good understanding of the design method. Considers alternative design concepts, and how the problems encountered in the design were solved.	5	5	5	5	5	5	5	5				
Content: Depth	No reference is made to literature or theory. The audience gain no new insights.	Explanations of concepts and/or theories are incomplete. Little attempt is made to tie theory to practice. The audience gains little from the presentation.	For the most part, explanations of concepts and theories are complete. Some helpful applications are included.	A complete explanation of major concepts and theories is provided and drawn upon relevant literature. Applications of theory are included to illuminate issues. Audience gain insights.	5	5	5	5	5	5	5	5				
Content: Accuracy	Content is sufficiently inaccurate. The audience may have been misled.	Enough errors are made but some information is accurate. The audience needs to determine what information is reliable.	No significant errors are made.	Content in the presentation is consistently accurate.	5	5	5	5	5	5	5	5				
Organization	Audience cannot understand presentation because there is no sequence of information.	Audience can follow presentation with effort. Some arguments are not clear. Organization seems random.	Presentation is generally clear and well organized. A few minor points are confusing.	Presentation is clear, logical and organized. Audience can follow line of reasoning.	5	5	5	5	5	5	5	5				
Professional Delivery	Presenters read the information to audience. Presenters are obviously anxious and cannot be heard. The audience is so distracted by the presenter's apparent difficulty with grammar and appropriate vocabulary that they cannot focus on the ideas presented.	Presenters seem uncomfortable and can be heard only if the audience is very attentive. Much of the information is read. Some grammatical errors and use of slang are evident. Some sentences are incomplete.	The presenters seem slightly uncomfortable at times, and the audience occasionally has trouble hearing him/her. For the most part, sentences are complete and grammatical, and they flow together easily.	Presenters are comfortable in front of audience and his/her voice is audible, No reading from the notes or presentation. Sentences are complete and grammatical, and they flow together easily.	5	5	5	5	5	5	5	5				

Rubric	Unacceptable (1)	Marginal (2)	Acceptable (3-4)	Exceptional (5)	Score of the Presenters											
					P1	P2	PC	G	P1	P2	PC	G	P1	P2	PC	G
Professional Visual Aids	The presentation is very much poorly prepared. • Font is too small • Too much and unimportant information is included. • Cluttered and too many misspellings.	The presentation is poorly prepared. • Font is too small • Too much information is included. • Cluttered and several misspellings	The presentation is well prepared. • Font is appropriate for reading • Appropriate information is included. • Uncluttered but a few misspellings.	The presentation is well prepared. • Font is large enough to be seen by all. • Information is organized to maximize audience understanding. • Uncluttered and no misspelling.	5	5	5	5	5	5	5	5				
Personal Appearance	Personal appearance (clothes, posture, ...) is very inappropriate	Personal appearance (clothes, posture,...) is somewhat appropriate	For the most part, personal appearance (clothes, posture, ... is appropriate	Personal appearance (clothes, posture,...) is very appropriate	5	5	5	5	5	5	5	5				
Team Work	No evidence of team work or collaboration	Some evidence of team work	The group worked as a team most of the time. Evidence of delineation of tasks	Teamwork was evident in all stages of the project. Strong evidence of delineation of tasks and team communication	5	5	5	5	5	5	5	5				
Results	Incorrect interpretation andlack of understanding of results	partial interpretation but incomplete understanding of results	Almost all the results have been discussed, minor improvements needed	All results have been interpreted and discussed	5	4	5	4	4	4	4	4				
Conclusion	Conclusion missing or misunderstood	Conclusion are rawn which are misstated	Important conclusion have been drawn but slight clarity needed	All important conclusions are made and clearly stated	4	4	4	4	4	4	4	4				
Total Score (out of 50)					49	48	49	48	48	48	48	48				

Declaration

	Name	Signature
Project Coordinator	K.Vijayan	

TitleoftheProject: Low Cost Digitalization (Industry 4.0) Solution for Siemens Sinumerik CNC System to Increase the Transparency and Utilization of the Machine

Presenter(s)Name: Pooja Anand (RA1511004010107)

(WriteReg.No.andNamefor eachstudentmember)

Supervisor(s)Name &Designation: Dr.P.Eswaran Associate professor

Date: 8/5/2019

Particulars	Unacceptable(1)	Marginal(2-3)	Acceptable(4)	Exceptional(5)	Score or N/A
Objective	Very little objective provided or information is incorrect	Some objective, but still missing some major points	Objective is nearly complete, missing some minor points	Objective complete and well-written; provides all necessary background principles for the experiment	5
Content	- Errors in technical content in many places - Contain little of the project details - An engineer would not be able to recreate the project based on the report.	- For the most part, technically correct - Contain a fair amount of technical details but incomplete - An engineer would have difficulty recreating the project based on the report.	- Technically correct - Contain most of the project details - An engineer might be able to recreate the project based on the report.	- Technically correct - Contain in-depth and completed details of the project. - An engineer can recreate the project based on the report.	5
Language (Word Choice, Grammar)	- Errors in sentence structure and grammar frequently distract the reader and interfere with meaning. - Unnecessary repetition of the same words and phrases. - Overuse of jargon and technical terms without definition. - Many misspelled words.	- In a few places, errors in sentence structure and grammar distract the reader and interfere with meaning. - Word choice could be improved. - Occasionally, technical jargon is used without definition. - A few misspelled words.	- For the most part, sentences are complete and grammatical, and flow together. Any errors are minor and do not distract the reader. - Repetition of words and phrases is mostly avoided. - For the most part, terms and jargon are used correctly with some attempt to define them. - One or two misspelled words.	- Sentences are complete and grammatical. They flow together easily - Words are chosen for their precise meaning. - Engineering terms and jargon are used correctly. - No misspelled words.	5
Experimental procedure	Missing several important experimental details or not written in paragraph format	Written in paragraph format, still missing some important experimental details	Written in paragraph format, important experimental details are covered, some minor details missing	Well-written in paragraph format, all experimental details are covered	5
Numerical Usage and Illustrations	- Figures, graphs, charts, and drawings are of poor quality, and have numerous inaccuracies and mislabeling, or may be missing. - No corresponding explanatory text for included items. - Inaccuracies in the equation. Little or no attempt is made to make it easy for the reader to understand the use of an equation or its derivation.	- In some cases, illustrations do not convey information clearly. - While items are labeled, references to these items are missing. - Most equations are accurate. Too many variables are not defined. Discussion regarding the development and usage of the equation is unclear.	- For the most part, illustrations are accurate, consistent with the text, and of good quality. - All items are generally labeled and are referred to in the text. - Most equations are accurate and clear. Most variables are defined and units specified. With some minor exceptions, an adequate discussion regarding the equation development and usage is stated.	- All figures, graphs, charts, and drawings are accurate, consistent with the text, and of good quality. They enhance understanding of the text. - All items are labeled and referred to in the text. - All equations are clear, accurate, and labeled. All variables are defined and units specified. Discussion about the equation development and use is stated.	5
Discussion	Very incomplete or incorrect interpretation of trends and comparison of data indicating a lack of understanding of results	Some of the results have been correctly interpreted and discussed; partial but incomplete understanding of results is still evident	Almost all of the results have been correctly interpreted and discussed, only minor improvements are needed	All important trends and data comparisons have been interpreted correctly and discussed, good understanding of results is conveyed	5
Conclusions	Conclusions missing or missing the important points	Conclusions regarding major points are drawn, but many are misstated, indicating a lack of understanding	All important conclusions have been drawn, could be better stated	All important conclusions have been clearly made, students show good understanding	5
Visual Format and Organization	- The document is not visually appealing. - There is no apparent ordering of paragraphs, and thus there is no progressive flow of ideas.	- Small errors are present - Within sections, the order in which ideas are presented is occasionally confusing.	- Structuring the content to represent the logical progression - The document is organized. - Use of white space helps the reader navigate the document, although the layout could be more effective.	- Structuring the content to represent the logical progression - The document is visually appealing and easily navigated. - Usage of white space is used as appropriate to separate blocks of text and add emphasis.	5
Use of references	- Little attempt is made to acknowledge the work of others. - Most references included are inaccurate or unclear.	- On several cases, references are not stated when appropriate. - References are not complete.	- With an occasional oversight, prior work is acknowledged. - With some minor exceptions, references are correct.	- Prior work is acknowledged by referring to sources for theories, assumptions, quotations, and findings. - Correct information for References.	4
Realistic constraints	Incorrect analysis on how this constraint affects the design of the system, component, or process.	Analysis contains a mixture of correct and incorrect reasons as to how this constraint affects the design of the system, component, or process.	Analysis provides correct reasons as to how this constraint affects the design of the system, component, or process but contains only a brief discussion.	Analysis provides correct reasons as to how this constraint affects the design of the system, component, or process and contains in-depth discussion.	5
				Total Score (out of 50)	49

K. Vijayan

Project coordinator

TitleoftheProject: Low Cost Digitalization (Industry 4.0) Solution for Siemens Sinumerik CNC System to Increase the Transparency and Utilization of the Machine

Presenter(s)Name: Vinitha Lea Philip (RA1511004010059)

(WriteReg.No.andNamefor eachstudentmember)

Supervisor(s)Name &Designation: Dr.P.Eswaran Associate professor

Date: 8/5/2019

Particulars	Unacceptable(1)	Marginal(2-3)	Acceptable(4)	Exceptional(5)	Score or N/A
Objective	Very little objective provided or information is incorrect	Some objective, but still missing some major points	Objective is nearly complete, missing some minor points	Objective complete and well-written; provides all necessary background principles for the experiment	5
Content	- Errors in technical content in many places - Contain little of the project details - An engineer would not be able to recreate the project based on the report.	- For the most part, technically correct - Contain a fair amount of technical details but incomplete - An engineer would have difficulty recreating the project based on the report.	- Technically correct - Contain most of the project details - An engineer might be able to recreate the project based on the report.	- Technically correct - Contain in-depth and completed details of the project. - An engineer can recreate the project based on the report.	5
Language (Word Choice, Grammar)	- Errors in sentence structure and grammar frequently distract the reader and interfere with meaning. - Unnecessary repetition of the same words and phrases. - Overuse of jargon and technical terms without definition. - Many misspelled words.	- In a few places, errors in sentence structure and grammar distract the reader and interfere with meaning. - Word choice could be improved. - Occasionally, technical jargon is used without definition. - A few misspelled words.	- For the most part, sentences are complete and grammatical, and flow together. Any errors are minor and do not distract the reader. - Repetition of words and phrases is mostly avoided. - For the most part, terms and jargon are used correctly with some attempt to define them. - One or two misspelled words.	- Sentences are complete and grammatical. They flow together easily - Words are chosen for their precise meaning. - Engineering terms and jargon are used correctly. - No misspelled words.	5
Experimental procedure	Missing several important experimental details or not written in paragraph format	Written in paragraph format, still missing some important experimental details	Written in paragraph format, important experimental details are covered, some minor details missing	Well-written in paragraph format, all experimental details are covered	5
Numerical Usage and Illustrations	- Figures, graphs, charts, and drawings are of poor quality, and have numerous inaccuracies and mislabeling, or may be missing. - No corresponding explanatory text for included items. - Inaccuracies in the equation. Little or no attempt is made to make it easy for the reader to understand the use of an equation or its derivation.	- In some cases, illustrations do not convey information clearly. - While items are labeled, references to these items are missing. - Most equations are accurate. Too many variables are not defined. Discussion regarding the development and usage of the equation is unclear.	- For the most part, illustrations are accurate, consistent with the text, and of good quality. - All items are generally labeled and are referred to in the text. - Most equations are accurate and clear. Most variables are defined and units specified. With some minor exceptions, an adequate discussion regarding the equation development and usage is stated.	- All figures, graphs, charts, and drawings are accurate, consistent with the text, and of good quality. They enhance understanding of the text. - All items are labeled and referred to in the text. - All equations are clear, accurate, and labeled. All variables are defined and units specified. Discussion about the equation development and use is stated.	5
Discussion	Very incomplete or incorrect interpretation of trends and comparison of data indicating a lack of understanding of results	Some of the results have been correctly interpreted and discussed; partial but incomplete understanding of results is still evident	Almost all of the results have been correctly interpreted and discussed, only minor improvements are needed	All important trends and data comparisons have been interpreted correctly and discussed, good understanding of results is conveyed	5
Conclusions	Conclusions missing or missing the important points	Conclusions regarding major points are drawn, but many are misstated, indicating a lack of understanding	All important conclusions have been drawn, could be better stated	All important conclusions have been clearly made, students show good understanding	5
Visual Format and Organization	- The document is not visually appealing. - There is no apparent ordering of paragraphs, and thus there is no progressive flow of ideas.	- Small errors are present - Within sections, the order in which ideas are presented is occasionally confusing.	- Structuring the content to represent the logical progression - The document is organized. - Use of white space helps the reader navigate the document, although the layout could be more effective.	- Structuring the content to represent the logical progression - The document is visually appealing and easily navigated. - Usage of white space is used as appropriate to separate blocks of text and add emphasis.	5
Use of references	- Little attempt is made to acknowledge the work of others. - Most references included are inaccurate or unclear.	- On several cases, references are not stated when appropriate. - References are not complete.	- With an occasional oversight, prior work is acknowledged. - With some minor exceptions, references are correct.	- Prior work is acknowledged by referring to sources for theories, assumptions, quotations, and findings. - Correct information for References.	4
Realistic constraints	Incorrect analysis on how this constraint affects the design of the system, component, or process.	Analysis contains a mixture of correct and incorrect reasons as to how this constraint affects the design of the system, component, or process.	Analysis provides correct reasons as to how this constraint affects the design of the system, component, or process but contains only a brief discussion.	Analysis provides correct reasons as to how this constraint affects the design of the system, component, or process and contains in-depth discussion.	4
				Total Score (out of 50)	48

K. Vijayan

Project coordinator

SRM Institute of Science and Technology
College of Engineering and Technology
Department of ECE

AY 2018-2019
15EC496L -Major Project Details (CO & PO Mapping)

Sl No	Register No	Students Name(s)	Project Supervisor	Project Title	Review 1 (10)	Review 2 (15)	Review 3 (20)
					CO1 & CO2	CO3 & CO4	CO5
					PO1, PO4, PO6, PO7	PO2, PO3, PO5, PO9	PO8, PO10, PO11, PO12
1	RA1511004010107	Pooja Anand	Dr. P. Eswaran	Low Cost Digitalization (Industry 4.0) Solution for Siemens Sinumerik CNC System to Increase the Transparency and Utilization of the Machine.	9.65	14.75	18.8
	RA1511004010059	Vinitha Lea Philip			9.3	14.75	18.7
2	RA1511004010553	kedar prasad karpe	DR. R .Kumar	cooperative transport using Multi-robot system	10	12.125	19.6
	RA1511004010511	Dhruv pant			10	12.25	19.6
	RA1511004010712	Nimish pastaria			10	12.25	19.6
	RA1511004010654	jayati singh			10	11.5	19.6

Course Outcomes:

- CO 1:** To provide learners with the opportunity to apply the knowledge and skills acquired in their courses to a specific problem or issue.
- CO 2:** To allow learners to extend their academic experience into areas of personal interest, working with new ideas, issues, organizations, and individuals.
- CO 3:** To encourage learners to think critically and creatively about academic, professional, or social issues and to further develop their analytical and ethical leadership skills.
- CO 4:** To provide learners with the opportunity to refine research skills and demonstrate their proficiency in written & oral communication skill.
- CO 5:** To take on the challenges of teamwork, prepare a presentation in a professional manner, and document all aspects of design work.

Program Outcomes

- PO 1: Engineering knowledge:
- PO 2: Problem analysis
- PO 3: Design/development of solutions:
- PO 4: Conduct investigations of complex problems
- PO 5: Modern tool usage
- PO 6: The engineer and society
- PO 7: Environment and sustainability
- PO 8: Ethics
- PO 9: Individual and team work
- PO 10: Communication
- PO 11: Project management and finance
- PO 12: Life-long learning



HOD/ECE



Coordinator

PROJECT REPORT – 5

6. TLP 5 for Review 1, 2, 3

FACULTY OF ENGINEERING AND TECHNOLOGY

SRM Institute of Science and Technology, Kattankulathur
(ACADEMIC YEAR 2018 - 2019 - EVEN)

FORMAT TLP5**Test Name : Review I****Component Max. Mark: 10.00 Marks****15EC496L(Major Project) handled by Mr.K.Vijayan(100233)**

S.No.	Reg. No	Name	Dept	Obtained Mark	%
1	RA1511004010030	M. Pranav	ECE	9.20	92.00
2	RA1511004010042	Srikanth S	ECE	9.50	95.00
3	RA1511004010059	Vinitha Lea Philip	ECE	9.30	93.00
4	RA1511004010095	Yvshanmuk Chowdary	ECE	6.50	65.00
5	RA1511004010107	Pooja Anand	ECE	9.65	96.50
6	RA1511004010119	P Sai Sisira	ECE	9.10	91.00
7	RA1511004010122	C.V.K. Anirudh Jagannath	ECE	9.30	93.00
8	RA1511004010128	Anurup Ojha	ECE	8.30	83.00
9	RA1511004010176	Akshat Singh	ECE	8.00	80.00
10	RA1511004010268	M Jagath	ECE	6.50	65.00
11	RA1511004010333	Deepansh Madan	ECE	7.50	75.00
12	RA1511004010482	Aditi Kothari	ECE	9.30	93.00
13	RA1511004010595	K Rohit	ECE	8.00	80.00

Total strength	13	Range of marks	No.of students
Total absentees	0	0-49	0
Total no. of failures	0	50-59	0
Pass MARK	50%	60-69	2
Pass percentage	100.00	70-79	1
		80-89	3
		90-100	7



SIGNATURE OF STAFF
Report Date:12-Sep-21



SIGNATURE OF HOD

FACULTY OF ENGINEERING AND TECHNOLOGY

SRM Institute of Science and Technology, Kattankulathur
(ACADEMIC YEAR 2018 - 2019 - EVEN)

FORMAT TLP5**Test Name : Review II****Component Max. Mark: 15.00 Marks****15EC496L(Major Project) handled by Mr.K.Vijayan(100233)**

S.No.	Reg. No	Name	Dept	Obtained Mark	%
1	RA1511004010030	M. Pranav	ECE	14.75	98.33
2	RA1511004010042	Srikanth S	ECE	14.75	98.33
3	RA1511004010059	Vinitha Lea Philip	ECE	14.75	98.33
4	RA1511004010095	Yvshanmuk Chowdary	ECE	12.50	83.33
5	RA1511004010107	Pooja Anand	ECE	14.75	98.33
6	RA1511004010119	P Sai Sisira	ECE	11.75	78.33
7	RA1511004010122	C.V.K. Anirudh Jagannath	ECE	15.00	100.00
8	RA1511004010128	Anurup Ojha	ECE	13.50	90.00
9	RA1511004010176	Akshat Singh	ECE	13.25	88.33
10	RA1511004010268	M Jagath	ECE	12.50	83.33
11	RA1511004010333	Deepansh Madan	ECE	13.00	86.67
12	RA1511004010482	Aditi Kothari	ECE	13.25	88.33
13	RA1511004010595	K Rohit	ECE	10.75	71.67

Total strength	13	Range of marks	No.of students
Total absentees	0	0-49	0
Total no. of failures	0	50-59	0
Pass MARK	50%	60-69	0
Pass percentage	100.00	70-79	2
		80-89	5
		90-100	6

K. Vijayan
SIGNATURE OF STAFF
Report Date:12-Sep-21

G. R.
SIGNATURE OF HOD

FACULTY OF ENGINEERING AND TECHNOLOGY

SRM Institute of Science and Technology, Kattankulathur
(ACADEMIC YEAR 2018 - 2019 - EVEN)

FORMAT TLP5**Test Name : Review III****Component Max. Mark: 20.00 Marks****15EC496L(Major Project) handled by Mr.K.Vijayan(100233)**

S.No.	Reg. No	Name	Dept	Obtained Mark	%
1	RA1511004010030	M. Pranav	ECE	19.10	95.50
2	RA1511004010042	Srikanth S	ECE	19.00	95.00
3	RA1511004010059	Vinitha Lea Philip	ECE	18.70	93.50
4	RA1511004010095	Yvshanmuk Chowdary	ECE	17.30	86.50
5	RA1511004010107	Pooja Anand	ECE	18.80	94.00
6	RA1511004010119	P Sai Sisira	ECE	17.70	88.50
7	RA1511004010122	C.V.K. Anirudh Jagannath	ECE	19.10	95.50
8	RA1511004010128	Anurup Ojha	ECE	17.80	89.00
9	RA1511004010176	Akshat Singh	ECE	18.50	92.50
10	RA1511004010268	M Jagath	ECE	17.40	87.00
11	RA1511004010333	Deepansh Madan	ECE	18.40	92.00
12	RA1511004010482	Aditi Kothari	ECE	18.50	92.50
13	RA1511004010595	K Rohit	ECE	18.70	93.50

Total strength	13	Range of marks	No.of students
Total absentees	0	0-49	0
Total no. of failures	0	50-59	0
Pass MARK	50%	60-69	0
Pass percentage	100.00	70-79	0
		80-89	4
		90-100	9



SIGNATURE OF STAFF
Report Date:12-Sep-21



SIGNATURE OF HOD

PROJECT REPORT – 5

7. Certificate by HoD



SRM
INSTITUTE OF SCIENCE & TECHNOLOGY
(Deemed to be University u/s 3 of UGC Act, 1956)

**SRM INSTITUTE OF SCIENCE AND TECHNOLOGY
COLLEGE OF ENGINEERING AND TECHNOLOGY
DEPARTMENT OF ELECTRONICS AND COMMUNICATION ENGINEERING**

CERTIFICATE

I hereby certify that the contents of the VIII semester
**15EC496L-Major Project (Sample Reports and Evaluation
Records)** are true to the best of my knowledge.

**Dr. Shanthi Prince
Head of the Department
Department of ECE**

Dr. Shanthi Prince
Professor and Head
Department of ECE
College of Engineering and Technology
SRM Institute of Science and Technology
Kattankulathur - 603 203, Chengalpattu District, Tamil Nadu